

1. MEASURES 1 (THEME FOUR)

- Converting bigger units to smaller units in length.
- Expressing smaller units to bigger units in length.
- Converting bigger units to smaller units in weight.
- Changing smaller units to bigger units in weight.
- Names of polygons
- Finding perimeter of regular polygons
- Calculating perimeter of a rectangle
- Finding one side to a regular polygon when perimeter is given.
- Finding one side of rectangle when perimeter is given.
- Finding area of rectangles
- Finding area of squares
- Finding one side of a square when area is given
- Finding area of triangles
- Finding area 67 separating figures
- Calculating area of shaded regions
- Finding volume and total surface area of cuboids
- Finding one side of cuboids

2. MEASURES 2 (THEME FOUR)

- The 12 hour –clock and 24 hours – clock time

- Converting 12 hour, clock time to 24 hour clock
- Changing 24 hour clock to 12 hour clock time
- Finding duration (length of time)
- Finding distance when speed and time is given
- Finding time when distance and speed is given
- Finding speed
- Simple rates and proportion
- Shopping bills and balances
- Transport charges
- Profit and loss
- Finding the cost price
- Finding the selling price

METRIC CONVERSION

We get the values of the metric conversion /units by comparing them with the basic or standard units.

Kilometre	Hecto	Deca	Basic	Deci	Centi	Milli
1000m	100m	10m	Metre Gram Litre	1/10m	1/100m	1/100m

Note

- a) The basic measure of length is metre
- b) The basic measure of weight is a gram
- c) The basic measure of capacity is a litre

Ordering of length = km Hm Dm M dm Cm mm

Ordering of weight = Kg Hg Dg G Dg Cg Mg

Ordering of capacity= Kl Hr Dl L Dl Cl MI

1 km = 1000m

1Hm	=	100m
1Dm	=	10m
1m	=	1m
1dm	=	0.1 or 1/10m
1cm	=	0.01 or 1/100m
1mm	=	0.001 Or 1/1000m
1kg	=	1000gm
1Hg	=	100gm
1Dg	=	10gm
1g	=	1gm
1dg	=	0.1gm
1cg	=	0.01gm
1mg	=	0.001gm

ACTIVITY

- 1KM = 1000METRES
- 1Hm = _____ metres
- 1Dm = _____ metres
- 1dm = _____ metres
- 1mm = _____ metres
- 1cm = _____ metres
- 1kg = _____ gms
- 1Hg = _____ gms
- 1 Dg = _____ gms
- 1dg = _____ gms
- How many metres make up 1 kilimetre?
- 1mg = _____ gms

CONVERTING KILOMETRES TO METRES

1. Express 12km as metres

$$\text{Since } 1\text{km} = 1000\text{m}$$

$$12\text{km} = 12 \times 1000$$

$$= 12000\text{m}$$

2. Convert 0.5km to metres

$$1\text{km} = 1000\text{m}$$

$$0.5\text{km} = (0.5\text{km} \times 1000) \text{ m}$$

$$= (5/10 \times 1000) \text{ m}$$

$$= 5 \times 100$$

$$= 500\text{m}$$

3. Express 12m as centimeters

$$1\text{m} = 100\text{cm}$$

$$12\text{m} = 12 \times 100$$

$$= 1200\text{cm}$$

4. Change 5 Hectometres to Deca metres

$$1\text{Hm} = 10\text{Dm}$$

$$5\text{Hm} = 5 \times 10$$

$$= 50\text{Dm}$$

ACTIVITY

Express the following as instructed.

1. 0.93km as metres
2. 6Hm as cm
3. $1 \frac{1}{2}$ km as Dm

4. 6Hm as metres
5. 9m as Cm
6. 0.9Km as Hm
7. $2\frac{1}{2}$ Km as metres
8. 1.2m as dm
9. 0.5dm as millimeters
10. 7Dm as mm
11. 2.5km as metres
12. 0.25m as cm

EXPRESSING SMALLER UNITS TO BIGGER UNITS IN LENGTH

1. Express 2000m to km

$$1000\text{m} = 1\text{km}$$

$$1\text{m} = \frac{1}{1000}\text{km}$$

$$2000\text{m} = \left(\frac{1}{1000} \times 2000\right) \text{ km}$$

$$= \frac{2000}{1000} \text{ km}$$

$$= 2 \text{ km}$$

2. Change 50cm to metres

$$100\text{cm} = 1\text{m}$$

$$1\text{cm} = \frac{1}{100}\text{m}$$

$$50\text{cm} = \left(\frac{1}{100} \times 50\right) \text{ m}$$

$$= \frac{50}{100} \text{ m}$$

$$= \frac{1}{2} \text{ m}$$

$$50\text{cm} = \frac{1}{2} \text{ m}$$

$$\text{or } 0.5\text{m}$$

3. Express 2500mm as decimeters

$$100\text{mm} = 1\text{dm}$$

$$1\text{mm} = \frac{1}{100}\text{dm}$$

$$2500\text{mm} = \frac{1}{100} \times 2500 \text{ dm}$$

$$\frac{1}{100} \times 2500$$

$$= 25\text{dm}$$

$$2500\text{mm} = 25\text{dm}$$

4. Convert 2750m to km

$$1000\text{m} = 1\text{km}$$

$$1\text{m} = \frac{1}{1000}\text{km}$$

$$2750\text{m} = \frac{1}{1000} \times 2750 \text{ km}$$

$$\frac{2750}{1000}$$

$$= \frac{2750}{1000}$$

$$\frac{275}{100}$$

$$\frac{275}{100}$$

$$= \frac{275}{100}$$

$$\frac{275}{100}$$

$$2750\text{m} = 2.75\text{km}$$

ACTIVITY

1. Express the following as instructed
2. 6000m as dm
3. 9000dm as cm
4. 2500m as Hm
5. 500m as km
6. 3400dm as Dm

CONVERSION OF BIGGER UNITS TO SMALLER UNITS IN WEIGHT

1. Express 2kg as grammes

$$1\text{kg} = 1000\text{g}$$

$$2\text{kg} = 2 \times 1000$$

$$= 2000\text{g}$$

2. Change $3\frac{1}{2}$ kg as grammes

$$1\text{kg} = 1000\text{g}$$

$$3\frac{1}{2}\text{kg} = 3\frac{1}{2} \times 1000$$

$$= \frac{3\frac{1}{2} \times 1000}{2}$$

$$= 7 \times 500$$

$$= 3500\text{g}$$

3. Change 0.5Hg to Dg

$$1\text{Hg} = 10\text{Dg}$$

$$0.5\text{Hg} = 0.5 \times 10\text{Dg}$$

$$= \frac{0.5 \times 10\text{Dg}}{10}$$

$$= 5 \times 1$$

$$0.5\text{Hg} = 5\text{Dg}$$

ACTIVITY

Change the following as instructed

1. 4kg to grammes
2. 0.5kg to grammes
3. $\frac{1}{2}$ Hg to grammes
4. $\frac{1}{4}$ g to mg
5. 3g to mg
6. 1kg to g
7. $2\frac{1}{2}$ kg to Dg
8. $1\frac{1}{2}$ kg to g
9. 0.4 Hg to dg
10. 0.25kg to g

CHANGING FROM SMALLER UNITS TO BIGGER UNITS IN WEIGHT

1. Express 400g as kg

$$1000\text{g} = 1\text{kg}$$

$$1\text{g} = \frac{1}{1000}\text{kg}$$

$$400\text{g} = \frac{1 \times 400}{1000}$$

$$= \frac{4}{\cancel{4000} \cancel{1000} 1}$$

$$400\text{g} = 0.4\text{kg}$$

2. Express 3700g as kg

$$100\text{g} = 1\text{kg}$$

$$1\text{g} = \frac{1}{1000}$$

$$3700\text{g} = \frac{(1 \times 3700) \text{ kg}}{1000}$$

$$= \frac{3700 \text{ kg}}{100}$$

$$= \frac{37 \text{ kg}}{10}$$

$$3700 = 3.7 \text{ kg}$$

ACTIVITY

Express the following grammes as kg

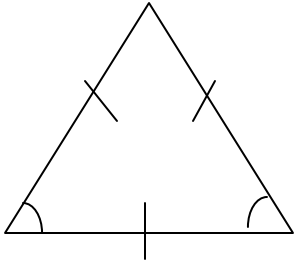
1. 3000g as kg
2. 1500dg as Hg
3. 200dg as g
4. 30000g as Dg
5. 7500g to kg
6. 250g to kg
7. 3400g as kg
8. 4400cg as Hg
9. 7000dg as Hg
10. 6700mg as g

NAMES OF POLYGONS

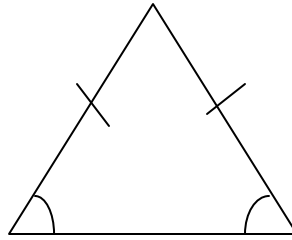
A polygon is any figure that is closed and has sides.

Any three sided polygon is called a triangle

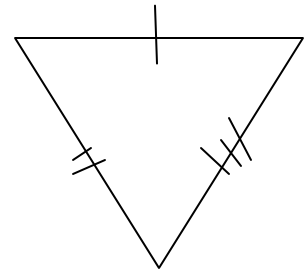
Examples



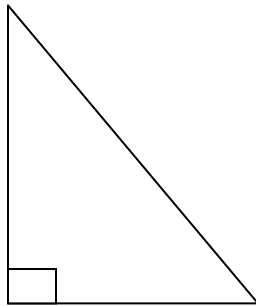
Equilateral triangle



Isosceles triangle

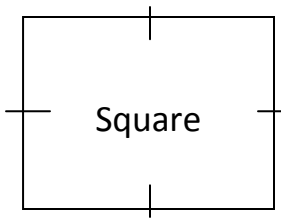


scalene triangle

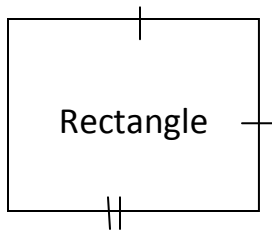


Right angled triangle

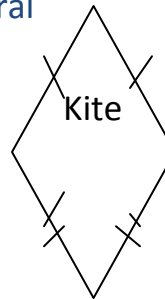
Any four sided polygon is called a quadrilateral



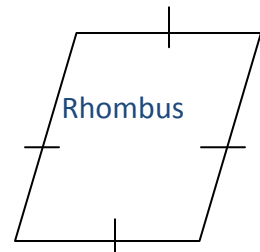
Square



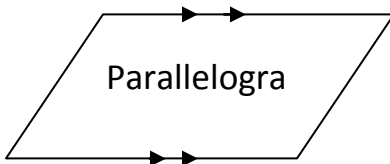
Rectangle



Kite



Rhombus



Parallelogram



Trapezium



Trapezium

- A five sided figure is a pentagon

- A six sided polygon is a hexagon
- A seven sided polygon is a Heptagon or Septagon
- An 8 sided polygon is an Octagon
- A 9 sided polygon is a nonagon
- A 10 sided polygon is a decagon
- An 11 sided polygon – Nuo-decagon
- A 12 sided polygon – Duo – decagon

ACTIVITY

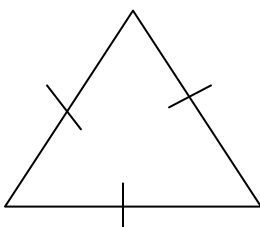
Name the following polygons

1. A 3 sided polygon is called _____
2. A six sided polygon is called _____
3. A 5 sided polygon is called _____
4. A 9 sided polygon is called _____
5. Draw and name these polygons
 - a) Equilateral triangle
 - b) Isosceles triangle
 - c) Trapezium
 - d) Pentagon
 - e) Kite

ENDING PERIMETER OF REGULAR POLYGONS

Find the perimeter of these regular polygons

1.



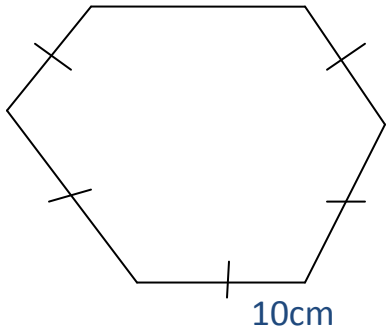
Perimeter = add all the sides

$$\text{Perimeter} = 5\text{cm} + 5\text{cm} + 5\text{cm}$$

Perimeter = 15cm

5cm

2.



Perimeter = sum of all sides

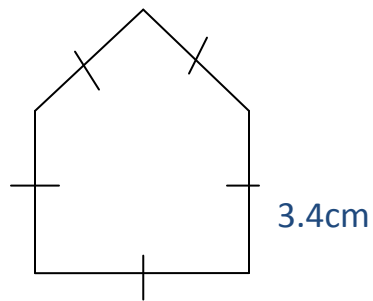
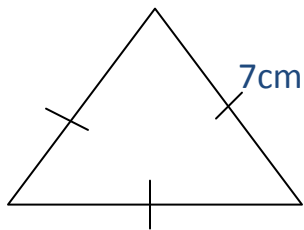
Perimeter = 10.5cm + 10.5cm + 10.5cm + 10.5cm
+ 10.5cm + 10.5cm + 10.5cm

P=63.0cm

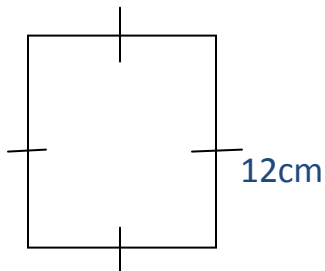
ACTIVITY

Find the perimeter of the following figures

1.



3.

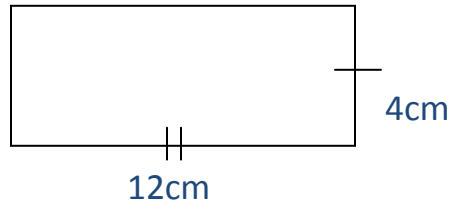


4. Find the perimeter of a regular pentagon with one side 10cm.
5. What is the perimeter of a regular octagon with a side 4.2cm
6. An equilateral triangle has each side 7cm. What is its perimeter?
7. One side of a regular hexagon measures 5cm. Find its perimeter.

FINDING PERIMETER OF A RECTANGLE

Perimeter is the way all round the polygon.

1. Find the perimeter of the rectangle below.



$$\text{Perimeter} = 2(L + W)$$

$$P = 2(12\text{cm} + 4\text{cm})$$

$$P = 2(16\text{cm})$$

$$P = 2 \times 16\text{cm}$$

$$P = 32\text{cm}$$

2. A rectangle measures 15cm long and 8cm wide. Calculate its perimeter.

$$\text{Perimeter} = 2(L + W)$$

$$P = 2(15\text{cm} + 8\text{cm})$$

$$P = 2(23\text{cm})$$

$$P = 2 \times 23\text{cm}$$

$$P = 46\text{cm}$$

3. **Calculate the perimeter of a rectangle**



$$\text{Perimeter} = 2(L + W)$$

$$P = 2(20\text{cm} + 8\text{cm})$$

$$P=2 \times (28\text{cm})$$

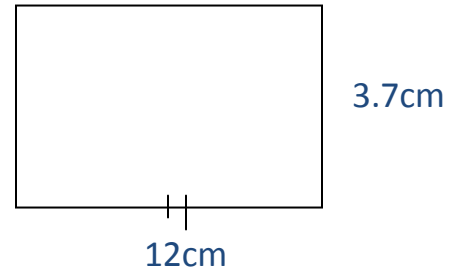
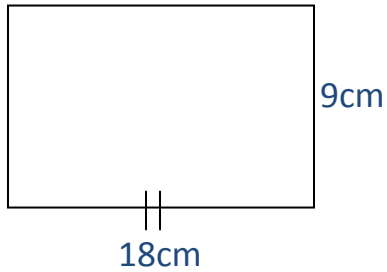
$$P=2 \times 28\text{cm}$$

$$P=56\text{cm}$$

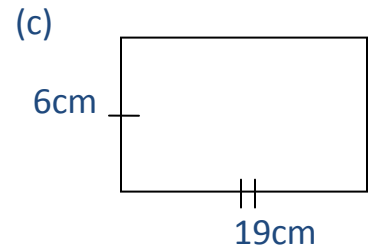
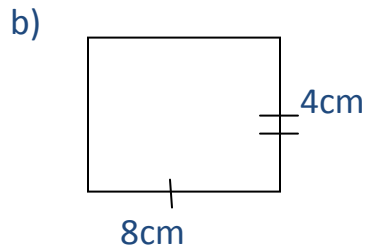
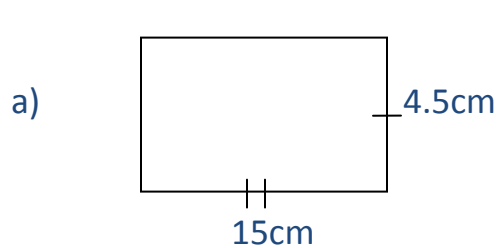
ACTIVITY

- Find the perimeter of the rectangles below.

a)



- A rectangle measures 15cm long by 12cm wide. Find its perimeter.
- Find the perimeter of the following figures.



- A rectangular field measures 40cm by 15cm. Calculate its perimeter.
- Find the perimeter of a rectangle whose length is 13cm and width 5cm.

FINDING ONE SIDE OF A REGULAR POLYGON WHEN PERIMETER IS GIVEN

- The perimeter of a square is 12cm. What is the length of each side?

A square has 4 sides

If 4 sides = 12cm

Let each side be L

$$L + L + L + L = 12\text{cm}$$

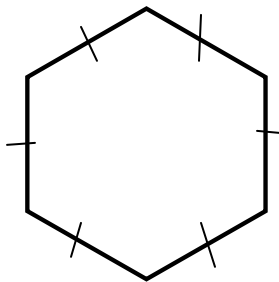
$$4L = 12\text{cm}$$

$$\begin{array}{r} 3 \\ \cancel{4}L = \cancel{12} \\ \cancel{4} \quad \cancel{4} \\ 1 \end{array}$$

$$L = 3\text{cm}$$

Each side is 3cm long

2. The perimeter of a polygon below is 24cm. How long is one of its sides?



If 6 sides = 24cm

Let each side be L

$$L+L+L+L+L+L = 24\text{cm}$$

$$6L = 24$$

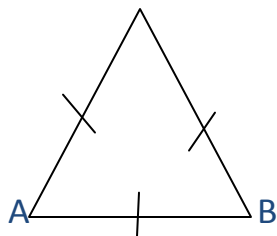
$$\begin{array}{r} 4 \\ \cancel{6}L = \cancel{24} \\ \cancel{6} \quad \cancel{6} \\ 1 \end{array}$$

$$L = 4\text{cm}$$

Each side is 4cm long

ACTIVITY

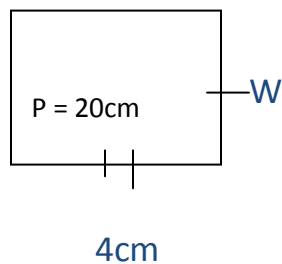
1. The polygon below has a perimeter of 15cm. Find the length of lien AB.



2. Find the side of a square whose perimeter is 64cm.
3. A pentagon has a perimeter of 100cm. How long is each of its sides if they are equal?
4. Find one side of a square whose perimeter is 138cm.
5. A rhombus has a perimeter of 32cm. How long is each of its sides?
6. The perimeter of a square is 112cm. Find the length of one side.
7. A regular pentagon has a perimeter of 225cm. Calculate the length of one side.

FINDING ONE SIDE OF RECTANGLE WHEN PERIMETER IS GIVEN

1. The perimeter of a rectangle is 20cm. If its length is 4cm. Find its width.



$P = \text{sum of all the sides}$

$$20\text{cm} = 4\text{cm} + 4\text{cm} + w + w$$

$$20\text{cm} = 8\text{cm} + 2w$$

$$20\text{cm} - 8\text{cm} = 2w$$

$$\frac{12\text{cm}}{2} = \frac{\cancel{2}w}{\cancel{2}}$$

$$6\text{cm} = w$$

$$\text{width} = 6\text{cm}$$

2. The perimeter of a rectangle is 30cm and its width is 5cm. Find its length.

$$P = 2(L + W)$$

$$30\text{cm} = 2(L + 5\text{cm})$$

$$30\text{cm} = 2L + 10\text{cm}$$

$$2L + 10\text{cm} = 30\text{cm}$$

$$2L + 10 - 10 = 30\text{cm} - 10\text{cm}$$

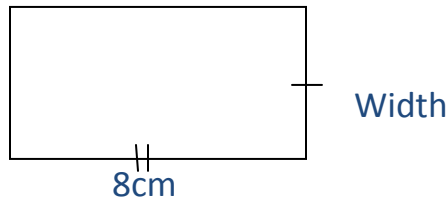
$$\frac{2L}{2} = \frac{20\text{cm}}{2}$$

$$L = 10\text{cm}$$

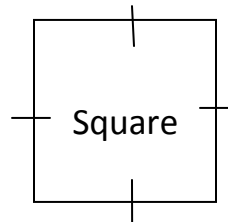
$$\text{Length} = 10\text{cm}$$

ACTIVITY

1. The perimeter of a rectangle is 20cm and its width is 3cm. Find its length.
2. The perimeter of a rectal is 28cm. Find its width in centimeters.



3. Find the width of a rectangle with a length of 20cm and perimeter of 60cm.
4. The perimeter of a square below is 100cm. Find the length of its side in centimeters.



5. Find the length of a rectangle with a width of 8cm and perimeter of 24cm.
6. The perimeter of a rectangle is 40cm. Its length is 15cm. Find the width.
7. The perimeter of a square is 60cm. Find one side in centimeters.

FINDING AREA OF RECTANGLES

A rectangle is a four sided polygon with two opposite sides.

The longer side of a rectangle is the length (L) and the shorter side is the width (W)

1. Find the area of the rectangle below.

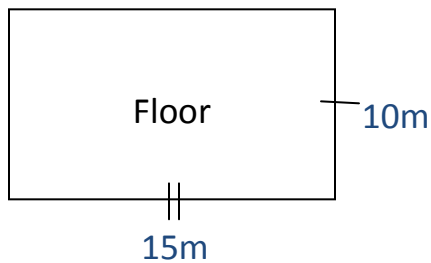


$$A = \text{Length} \times \text{width}$$

$$A = 7\text{cm} \times 4\text{cm}$$

$$A = 28 \text{ cm}^2$$

2. Find the area of the rectangular floor whose length is 15m and width 10m.



$$A = \text{Length} \times \text{width}$$

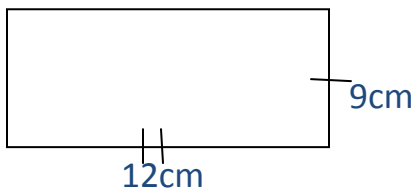
$$A = 15\text{m} \times 10\text{m}$$

$$A = 150\text{m}^2$$

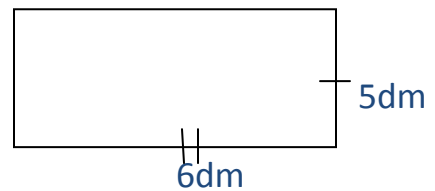
ACTIVITY

Find the area of the rectangles below

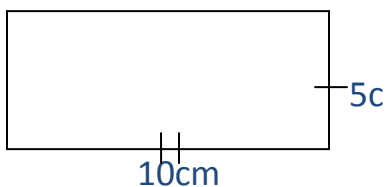
1.



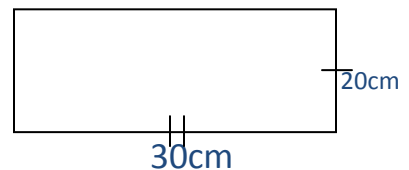
2.



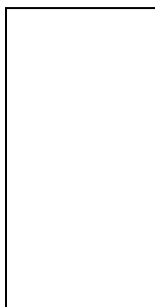
3.



4.



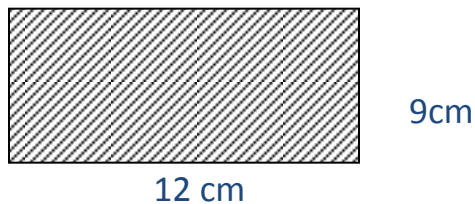
5.



—8cm
—

|
4cm

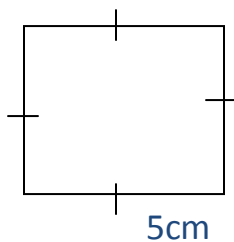
6. Find the area of a rectangle whose length is 6m and width 4.5cm.
7. A rectangular piece of land measures 48m by 30m. What is its area?
8. A rectangular piece of cloth is 40 metres long and 3 metres wide. What is its area?
9. Find the area of the rectangle below.



10. A football field measures 150m by 60m. Find its area.

FINDING AREAS OF SQUARES

1. Find the area of a square below.



Area = side x side

Area = 5 x 5 (all sides are equal)

Area = 5cm x 5cm

Area = 25cm^2

2. A square board has a side of 3cm

Find its area

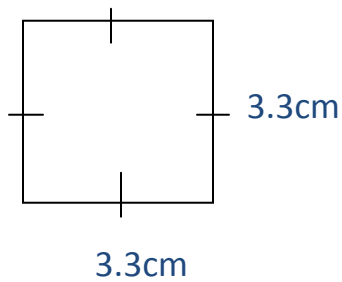
$$\text{Area} = \text{side} \times \text{side}$$

$$\text{Area} = S \times S$$

$$\text{Area} = 3\text{cm} \times 3\text{cm}$$

$$\text{Area} = 9\text{cm}^2$$

3. Find the area of a square below.



$$\text{Area} = \text{side} \times \text{side}$$

$$\text{Area} = 5 \times 5$$

$$\text{Area} = 3.3.\text{cm} \times 3.3\text{cm}$$

$$\text{Area} = 3.3$$

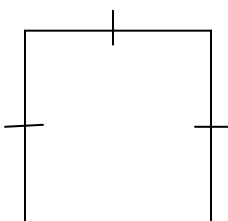
$$\begin{array}{r} \text{x}3.3 \\ 99 \\ +99 \\ \hline 10.89 \end{array}$$

$$A = 10.89\text{cm}^2$$

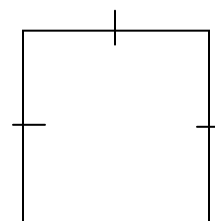
ACTIVITY

Find the area of the squares below

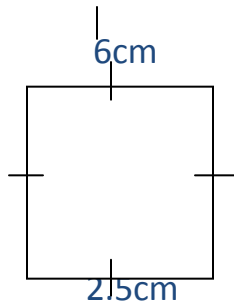
- 1.



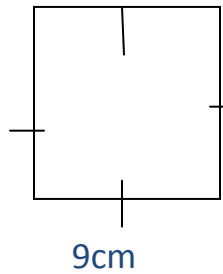
- 2.



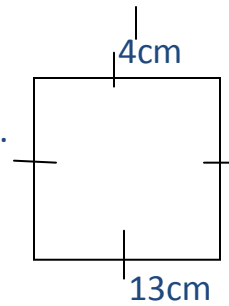
3.



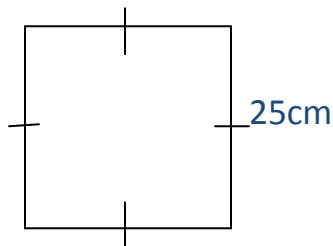
4.



5.



6. Find the area of a square mat which is 4.5cm by 4.5cm
7. A square garden has a length of 55cm. Calculate its area.
8. A square measures 35cm by 35cm. Find its area.
9. A square room measures 40m by 40m. Find the area of the carpet that covers the room exactly.
10. Find the area of the square below.



FINDING ONE SIDE OF A SQUARE WHEN AREA IS GIVEN

1. A square has an area of 25cm^2 . Find the size of each side.

$$\text{Area} = \text{side} \times \text{side}$$

$$A = S \times S$$

$$25\text{cm}^2 = S \times s$$

$$25 = S^2$$

$$\sqrt{25} = \sqrt{S^2}$$

$$\sqrt{5 \times 5} = \sqrt{S \times S}$$

$$5 = \text{side}$$

side work

5	25	
5	5	
	1	

$$\sqrt{25} = 5$$

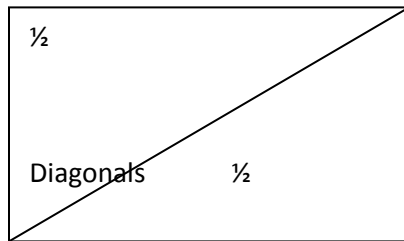
each side = 5cm

ACTIVITY

1. Find the side of a square with an area of 16cm^2
2. Find the side of a square with an area of 48cm^2
3. Work out the area of a square that has an area of 36cm^2
4. A square has an area of 64cm^2 . Find the size of one side.
5. A square has an area of $8/\text{cm}^2$. Find length of each side.
6. Find the length of a square whose area is 100cm^2 .
7. Calculate the length of a square of area 144cm^2 .

FINDING AREA OF TRIANGLES

If you cut a rectangle through the diagonal get 2 equal triangles.



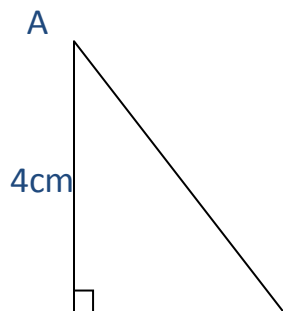
Area of the shaded triangle is $\frac{1}{2}$ the area of the rectangle.

The area of triangle is $\frac{1}{2} \times L \times W$

We call the length and the width of a triangle a base (b) and a height (h)

A sign of a right angle $\perp$ is often used to touch the base and the height.

EXAMPLES



B 3cm C

In this triangle,

BC is the base = 3cm

AB is the height = 4cm

Area = $\frac{1}{2} \times \text{base} \times \text{height}$

Area = $\frac{1}{2} bh$

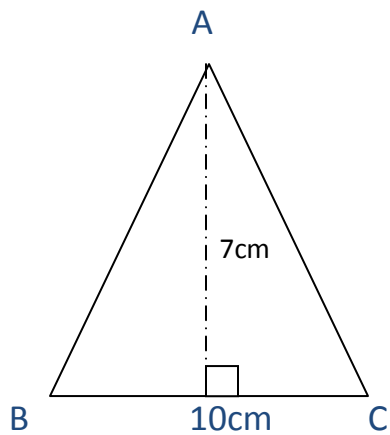
Area = $\frac{1}{2} \times 3\text{cm} \times 4\text{cm}$

$$\text{Area} = \frac{1}{2} \times 3\text{cm} \times 4\text{cm}$$

$$\text{Area} = 3 \times 2$$

$$\text{Area} = 6\text{cm}^2$$

2.



In this triangle

BC is the base = 10cm

AO is the height = 7cm

Area = $\frac{1}{2} \times \text{base} \times \text{height}$

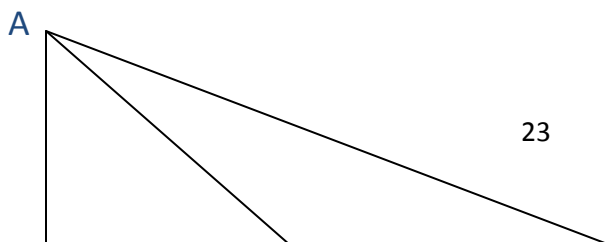
$$A = \frac{1}{2} \times b \times h \quad 5$$

$$A = \frac{1}{2} \times 10\text{cm} \times 7\text{cm}$$

$$A = 5\text{cm} \times 7\text{cm}$$

$$A = 35\text{cm}^2$$

3.



height

6cm

O 4cm B 5cm C

Area of the shaded part

In this triangle

BC is the base = 5cm

AO is the height = 6cm

Area = $\frac{1}{2}$ x base x height

A = $\frac{1}{2}$ x b x h

$$\frac{1}{2} \times 5\text{cm} \times 6\text{cm}$$

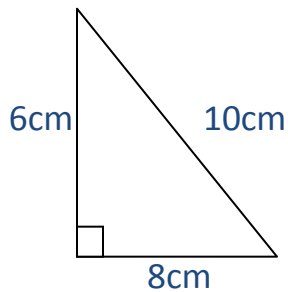
$$= 5\text{cm} \times 3\text{cm}$$

$$A = 15\text{cm}^2$$

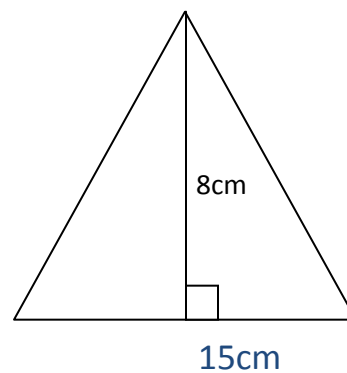
ACTIVITY

Find the area of the triangle below.

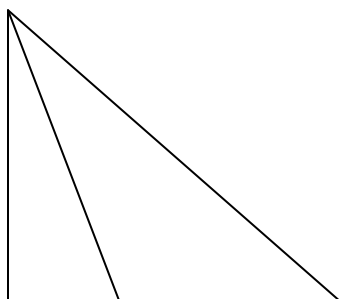
1.



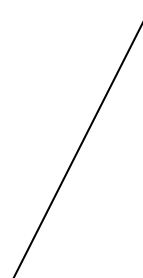
2.



3.



4.



13cm

5cm

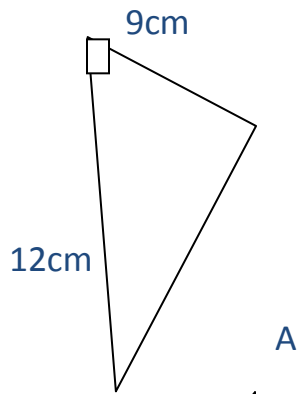
10cm

5cm

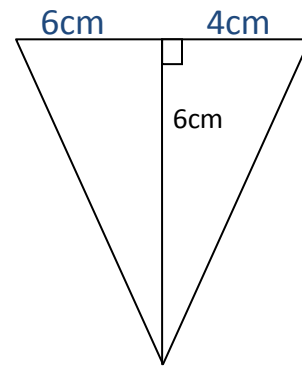
10cm

12cm

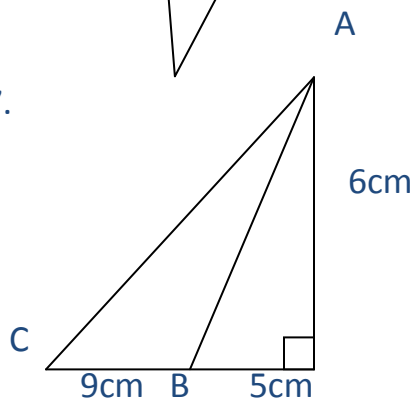
5.



6.



7.



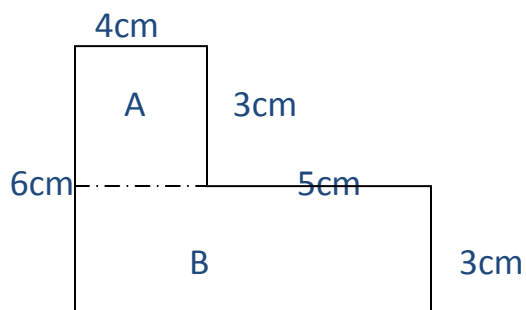
8. Find the area of a triangle with a base of 3cm and height 10cm.

9. The area of a triangle is 24cm^2 . Its base is 8cm. Find its height.

10. Find the area of a triangle of 24cm base and height 18cm

FINDING AREA BY SEPARATING FIGURES

1. Find the area of the figure below.



9cm

Area of A = L x W

$$A = 4\text{cm} \times 3\text{cm}$$

$$A = 12\text{cm}^2$$

Area of B = L x W

$$A = (9 \times 3) \text{ cm}$$

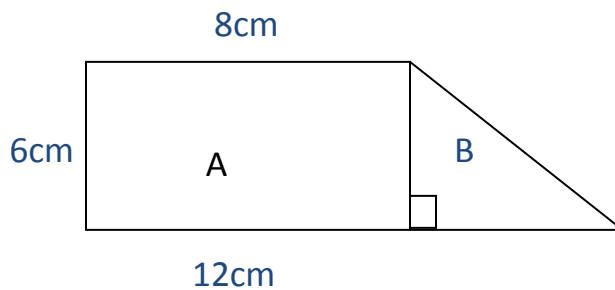
$$A = 27\text{cm}^2$$

Area of figure = Area of A + area of B

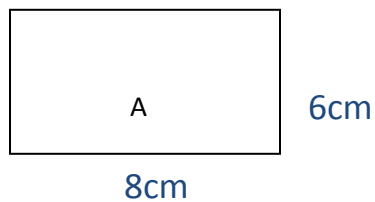
$$= 12\text{cm}^2 + 27\text{cm}^2$$

$$= 39\text{cm}^2$$

2. Find the area of the figure

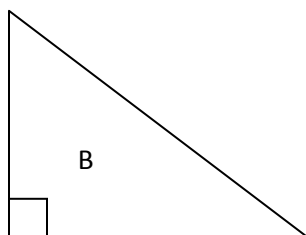


Area of rectangle = L x W



$$8\text{cm} \times 6\text{cm}$$

$$48\text{cm}^2$$



Area of triangle

6cm

4cm

$$A = \frac{1}{2} \times b \times h$$

$$A = \frac{1}{2} \times 4\text{cm} \times 6\text{cm}$$

$$A = 2\text{cm} \times 6\text{cm}$$

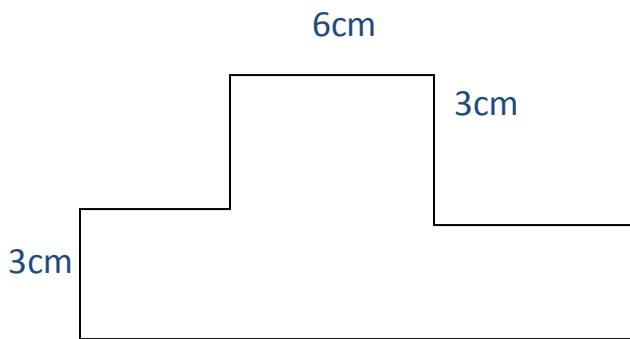
$$A = 12\text{cm}^2$$

Area of A + area of B

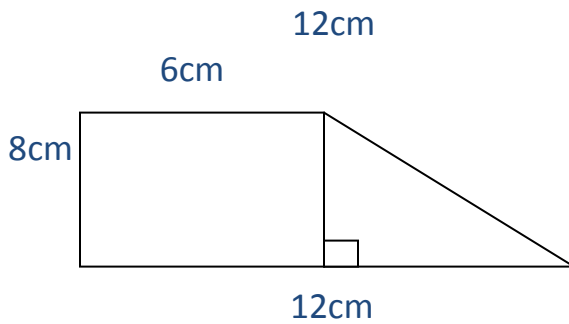
$$= 48\text{cm}^2 + 12\text{cm}^2 = 60\text{cm}^2$$

Find area of the following figures

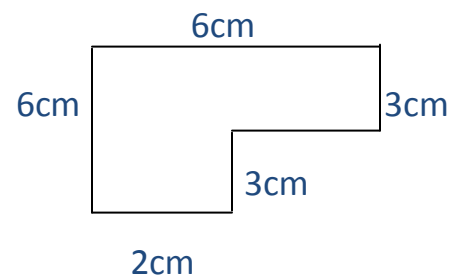
1.



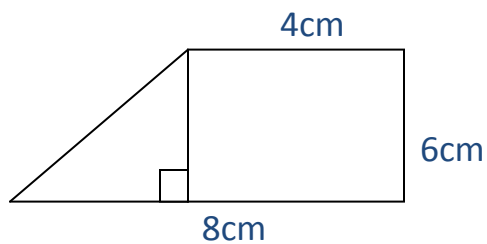
2.



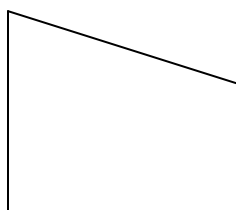
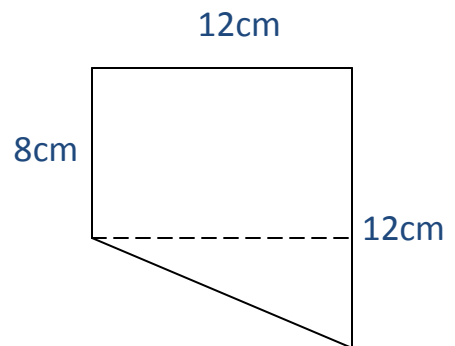
3.

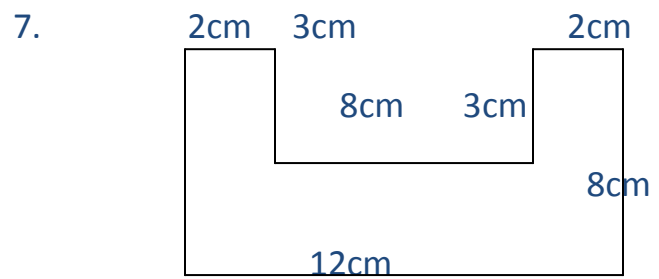


4.



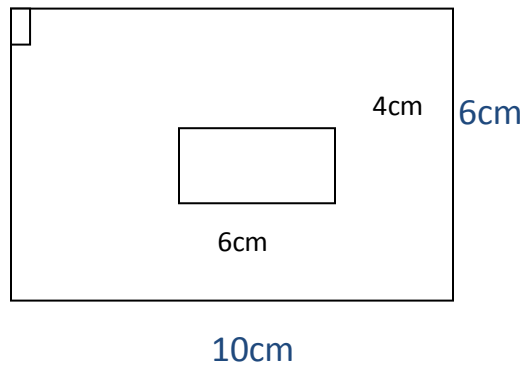
5.





CALCULATING AREA OF SHADED AND UNSHADED REGIONS

1. Find the area of the shaded part.



Area of a big rectangle

$$A = L \times W$$

$$A = 10\text{cm} \times 6\text{cm}$$

$$A = 60\text{cm}^2$$

Area of a small rectangle

$$A = L \times W = 6\text{cm} \times 4\text{cm}$$

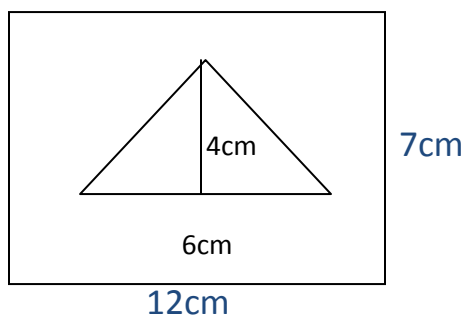
$$A = 24\text{cm}^2$$

Area of shaded region = Area of big rectangle minus area of small rectangle.

$$= 60\text{cm}^2 - 24\text{cm}^2$$

$$= 36\text{cm}^2$$

- 2.



Area of rectangle

$$A = L \times W$$

$$A = 12\text{cm} \times 7\text{cm}$$

$$A = 84\text{cm}^2$$

Area of triangle

$$A = \frac{1}{2} \times b \times h$$

Area of shaded part

$$A = \frac{1}{2} \times 6\text{cm} \times 4\text{cm}$$

$$A = 12\text{cm}^2$$

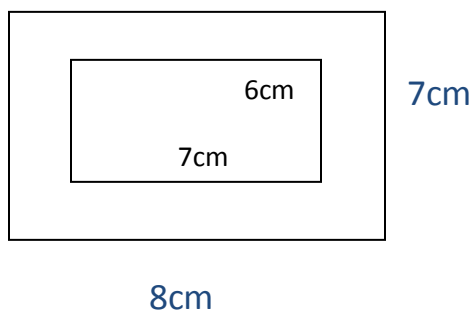
$$= 84\text{cm}^2 - 12\text{cm}^2$$

$$A = 72\text{cm}^2$$

ACTIVITY

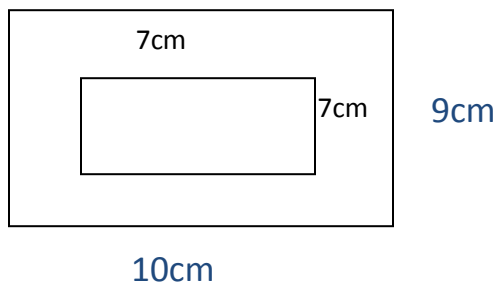
Find the area of the shaded part in the figures.

1.

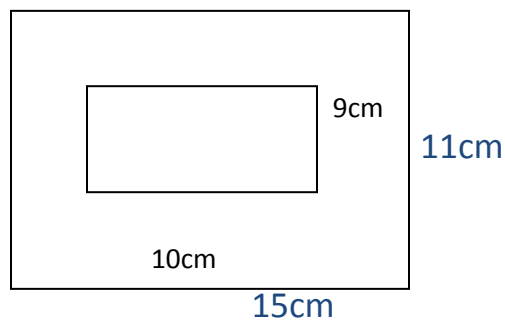


Find the area of un shaded part

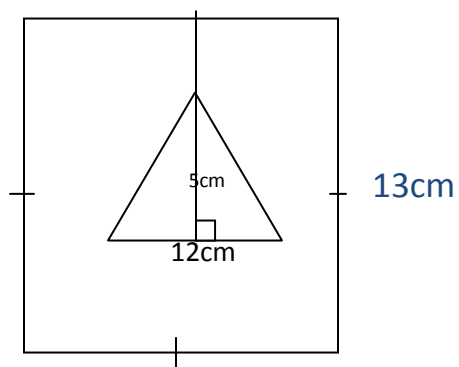
2.



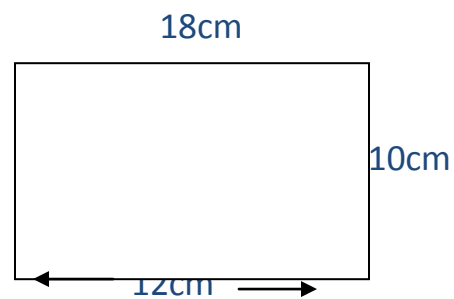
3.



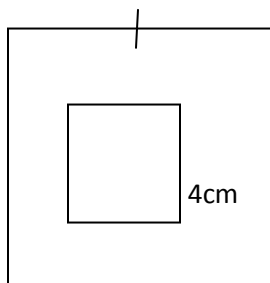
4.



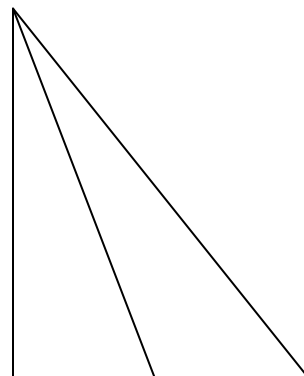
5.



6.

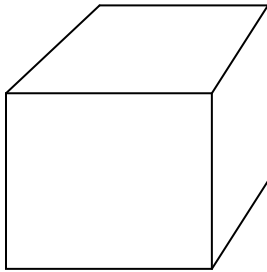


7.





1. FINDING VOLUME

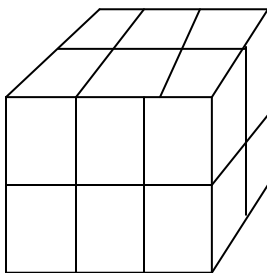


Cuboid

Find the number of:

- i) Vertices - 8
- ii) Edges - 12
- iii) Faces - 6

2.



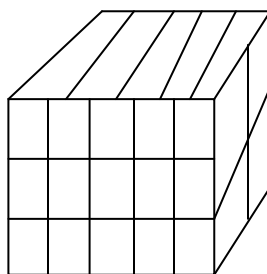
2 a long height

2 along width

3 along length

$$= 3 \times 2 \times 2 = 12 \text{ cubic units}$$

$$\text{volume} = 12 \text{ cubic units}$$



5 cubes

2 cubes

2 cubes

Length = 5 cubes

Width = 2 cubes

Height = 2 cubes

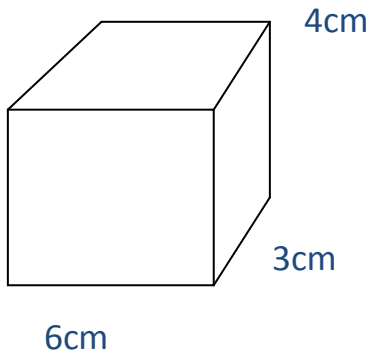
$$\text{Volume} = L \times w \times h$$

$$\text{Volume} = 5\text{cm} \times 2\text{cm} \times 2\text{cm}$$

$$\text{Volume} = 10\text{cm}^2 \times 2\text{cm}$$

$$\underline{\text{Volume} = 20\text{cm}^3}$$

It can also be written as 20 cubic cm



- a) Find the volume of the figure.

Volume = Length x width x height

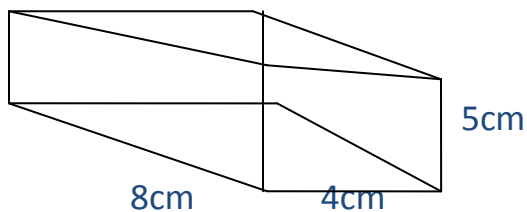
Volume = $L \times w \times h$

Volume = $6\text{cm} \times 3\text{cm} \times 4\text{cm}$

Volume = $6\text{cm} \times 12\text{cm}^2$

Volume = 72cm^3

2. Calculate the volume of the figure



$$\text{Volume} = L \times w \times h$$

$$\text{Volume} = 8\text{cm} \times 4\text{cm} \times 5\text{cm}$$

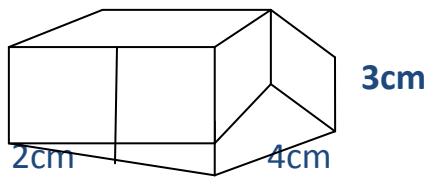
$$\text{Volume} = 8\text{cm} \times 20\text{cm}$$

$$= 160\text{cm}^3$$

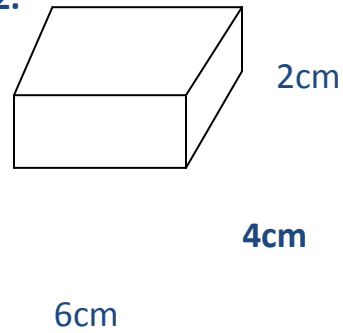
ACTIVITY

Find the volume

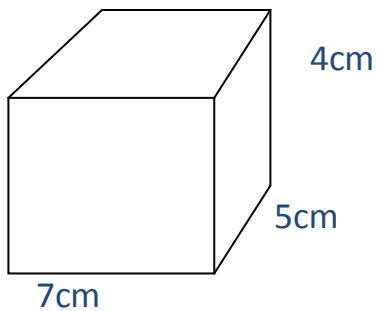
1.



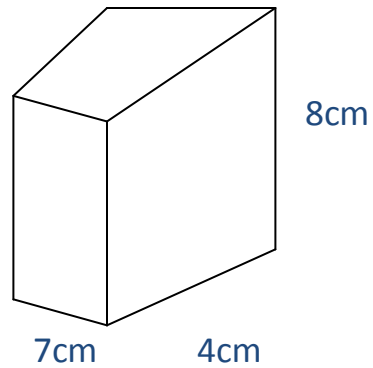
2.



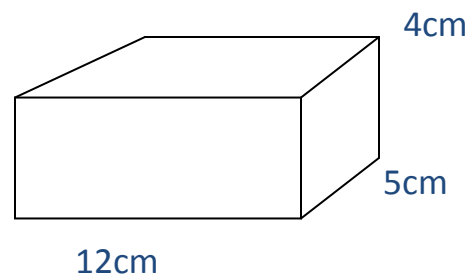
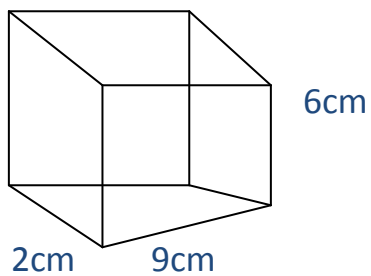
3.



4.



5.

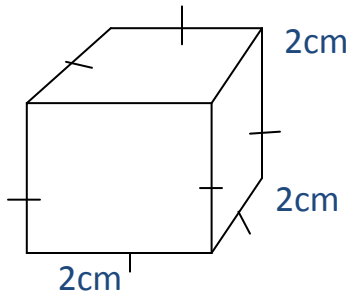


7. Find the volume of a box whose length is 30cm, width is 10cm and height is 2cm.
8. A box measures 15cm by 8cm by 2cm. Calculate the volume of the box.

9. Find the volume of a box measuring 20cm by 6cm by 4cm.

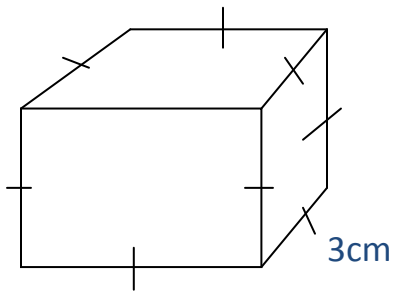
VOLUME OF CUBES

A cube has all its faces equal. Each face is a square.



All sides are equal

1. Find the volume of a cube.



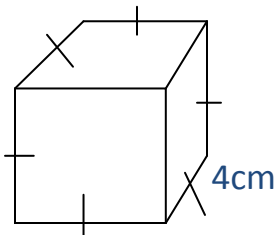
$$\text{Volume} = S \times S \times S$$

$$\text{Volume} = 3\text{cm} \times 3\text{cm} \times 3\text{cm}$$

$$\text{Volume} = 9 \times 3$$

$$= 27\text{cm}^3$$

2. Find the volume of a cube below.



$$\text{Volume} = L \times L \times L$$

$$\text{Volume} = 4\text{cm} \times 4\text{cm} \times 4\text{cm}$$

$$\text{Volume} = 64\text{cm}^3$$

ACTIVITY

Find the volume of a cube whose side is;

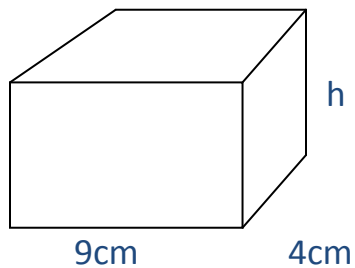
1. 5cm
2. 6cm

3. 7dm
4. 8m
5. 3dm
6. 0.2cm
7. 10cm
8. 0.5m
9. 2.3cm
10. $3\frac{1}{2}$ cm

FINDING ONE SIDE OF ACUBOID WHEN GIVEN VOLUME AND TWO SIDES

Example

1. Find the height of acuboid below whose volume is 180cm^3 .



Volume = length x width x height

$$180\text{cm}^3 = 9\text{cm} \times 4\text{cm} \times h$$

$$180\text{cm}^3 = 36\text{cm}^2 h$$

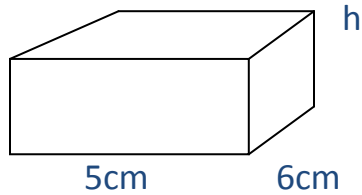
$$\frac{180\text{cm}^3}{36\text{cm}^2} = \frac{36\text{cm}^2 h}{36\text{cm}^2}$$

$$5\text{cm} = h$$

height = 5cm

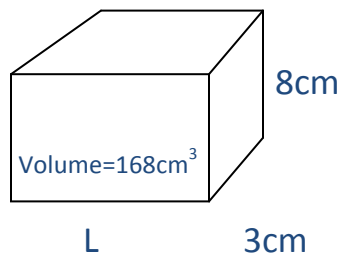
ACTIVITY

1. Find the height of the cuboid if its volume is 120cm^3 .

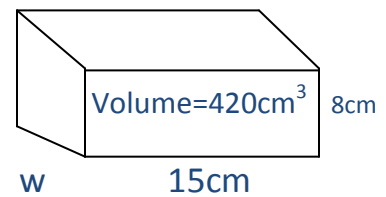


2. Find the unknown sides in the figures below.

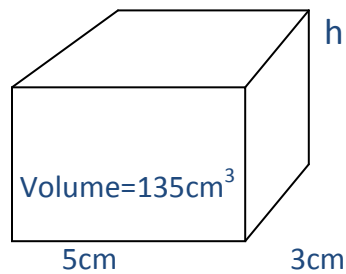
a)



(b)



c)

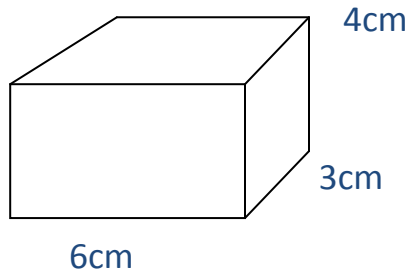


3. The volume of the cuboid is 60cm^3 . Find the width of the cuboid if its length and height are 5cm and 3cm respectively.
4. Given the volume of a cuboid is 72cm^3 . Find the height of the cuboid if its length is 8cm and width 3cm

TOTAL SURFACE AREA (TSA) OF CUBOIDS AND CUBES

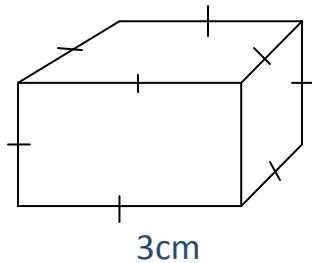
Examples

1. Find the total surface area of the figure below.



$$\begin{aligned}\text{Total surface area (TSA)} &= 2(L \times w) + 2(w \times h) + 2(L \times h) \\ &= 2(6\text{cm} \times 3\text{cm}) + 2(3\text{cm} \times 4\text{cm}) + 2(6\text{cm} \times 4\text{cm}) \\ &= 2(18\text{cm}^2) + 2(12\text{cm}^2) + 2(24\text{cm}^2) \\ &= 36\text{cm}^2 + 24\text{cm}^2 + 48\text{cm}^2 \\ &= 108\text{cm}^2\end{aligned}$$

2. Find the total surface area of the cube below.

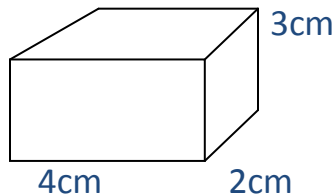


$$\begin{aligned}\text{T.S.A} &= 6(5 \times 5) \\ &= 6(3\text{cm} \times 3\text{cm}) \\ &= 6 \times 9\text{cm}^2 \\ &= 54\text{cm}^2\end{aligned}$$

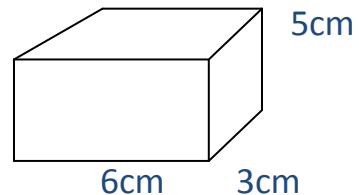
ACTIVITY

Find the total surface area of the following.

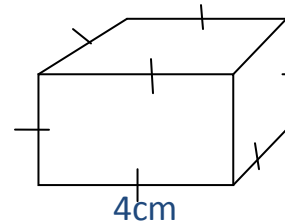
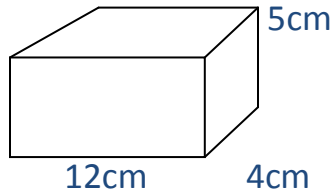
1.



2.



3.



5. Find the TSA of a box whose length is 9cm, width 10cm and height 3cm.
7. Find the TSA of a cube measuring 5cm.

THE 12 – HOUR CLOCK AND THE 24 HOUR CLOCK TIME

NOTE:

A new day begins just after midnight and ends at midnight.

One day has 24 hours.

a.m means ANTEMERIDIEM (Morning time)

p.m means POST MERIDIEM (afternoon or evening time)

TELLING TIME USING a.m and p.m

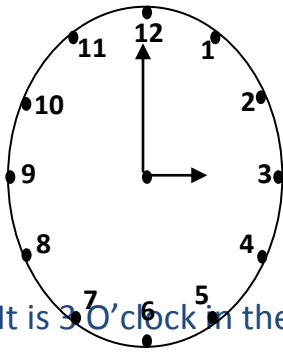
We use a.m after midnight and in the morning and p.m in the afternoon and evening.

Examples

Write the time shown on the clock face.

Evening

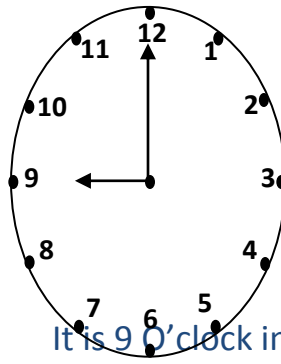
Morning



It is 3 O'clock in the evening or

3:00p.m

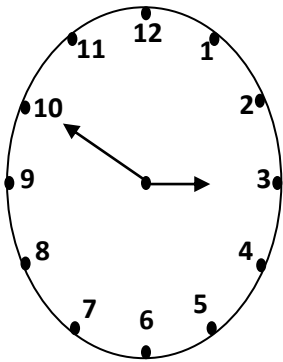
Afternoon



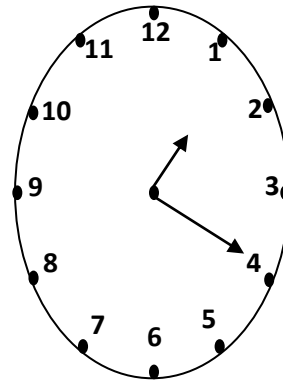
It is 9 O'clock in the morning or

9:00a.m

Afternoon



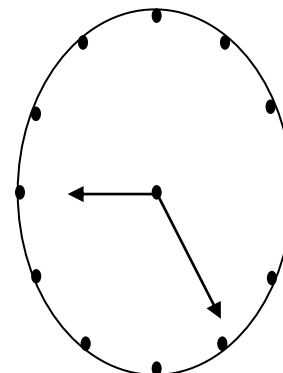
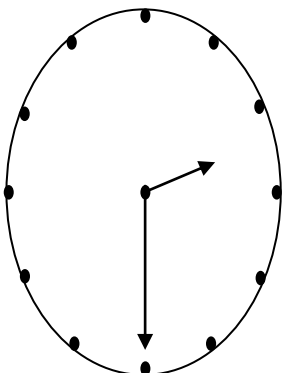
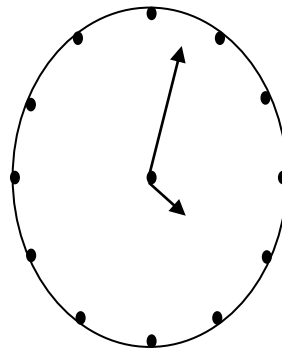
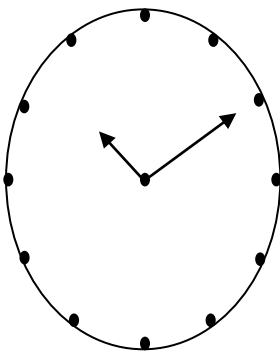
It is 10 minutes to 3 O'clock in the afternoon or 2:50pm.

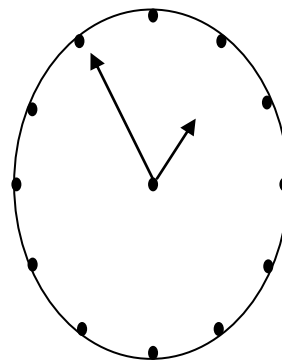
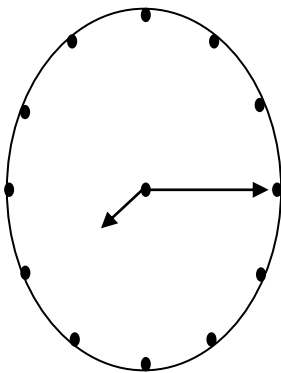


It is 20 minutes past 1 O'clock in the afternoon or 1:20pm.

ACTIVITY

Write the time on the clock face using a.m or p.m





**COMPARISING 12 HOUR CLOCK TIME AS a.m AND p.m WITH 24 HOUR
CLOCK TIME GIVEN IN HORUS**

12 HOUR CLOCK	24 HOUR CLOCK
12 midnight	24.00hrs (Full day)
11.00pm	23.00hrs
10.00p.m	22.00hrs
9.00p.m	21.00hrs
8.00p.m	20.00hrs
7.00p.m	19.00hrs
6.00p.m	18.00hrs
5.00p.m	17.00hrs
4.00p.m	16.00hrs
3.00p.m	15.00hrs
2.00p.m	14.00hrs
1.00p.m	13.00hrs
12.00 mid-day	12.00hrs (1/2 day)
11.00a.m	11.00hrs
10.00a.m	10.00hrs
9.00a.m	09.00hrs

8.00a.m	08.00hrs
7.00a.m	07.00hrs
6.00a.m	06.00hrs (sun raising)
5.00a.m	05.00hrs
4.00a.m	04.00hrs
3.00a.m	03.00hrs
2.00a.m	02.00hrs
1.00a.m	01.00hrs
12.00midnight	00.00hrs(New day starts)

CHANGING 12 HOUR CLOCK TO 24 HOUR CLOCK TIME

Express 3.45a.m in 24hour clock time.

$$\begin{array}{r}
 = \quad 3.45 \\
 + \underline{00.00} \\
 \hline
 03.45 \qquad = 03.45\text{hrs}
 \end{array}$$

3.45am = 03.45hrs.

When time is given in a.m, we add 00.00hrs

2. Change 7.00p.m to 24 hour clock

$$\begin{array}{r}
 = \quad 7.00\text{p.m} \\
 + \underline{12.00} \\
 \hline
 19.00\text{hrs}
 \end{array}
 \quad \longrightarrow \quad 7.00\text{pm} = 19.00\text{hrs}$$

When time is given in p.m, we add 12.00hrs

3. Change 2.30a.m to 24hr clock.

$$\begin{array}{r}
 = \quad 2.30 \\
 + \underline{00.00} \\
 \hline
 \underline{02.30} \quad 02.30\text{hrs}
 \end{array}$$

4. Change 9.40p.m to 24hr clock

$$\begin{array}{r}
 = \quad 9.40\text{pm} \\
 + \underline{12.00}
 \end{array}$$

$$\underline{21.40} = 21.40\text{hrs}$$

5. Change 07.30a.m to 24hr clock

$$= \begin{array}{r} 7.30\text{am} \\ +00.00 \\ \hline 07.30 \end{array} = 07.30\text{hrs}$$

ACTIVITY

Change the following to 24hr clock.

1. 1.00a.m
2. 3.00a.m
3. 7.00a.m
4. 9.00a.m
5. 12.00 Noon
6. 2.00p.m
7. 4.00p.m
8. 6.00p.m
9. 10.00p.m
10. 1.30p.m

CHANGING 24 HOUR CLOCK TO 12 HOUR CLOCK

1. Write 02.00hrs in the hour clock time.

$$= \begin{array}{r} 02.00\text{hrs} \\ - \underline{00.00} \\ \hline 02.00 \end{array} = \underline{2.00\text{hrs} = 2.00\text{a.m}}$$

2. Change 15.00hrs to a 12 hour clock time.

$$= 15.00\text{hrs}$$

$$\begin{array}{rcl}
 & - 12.00 & \\
 03.00 & = & 3.00\text{hr} \\
 \hline
 15.00\text{hrs} & = & 3.00\text{pm}
 \end{array}$$

3. Change 07.30hrs to the 12 hour clock time

$$\begin{array}{rcl}
 = & 07.30\text{hrs} & \\
 & - 00.00 & \\
 \hline
 & 7.30 & = 7.30\text{a.m}
 \end{array}$$

4. Change 23.10hrs in the 12 hour clock time

$$\begin{array}{rcl}
 = & 23.10 & \\
 & - 12.00 & \\
 \hline
 & 11.10 & = 11.10\text{pm} \\
 23.10\text{hrs} & = & 11.10\text{p.m}
 \end{array}$$

ACTIVITY

Change to 12 hour clock

1. 01.00hrs
2. 03.00hrs
3. 09.00hrs
4. 07.00hrs
5. 10.00hrs
6. 11.00hrs
7. 12.00hrs
8. 16.00hrs
9. 24.00
10. 02.15hrs

FINDING DURATION (Length of time)

1. The bus started its journey to Mbale at 9.00a.m and reached the town at 1.30p.m. How long did the journey take?

hrs	:	minutes	
12	:	00noon	
<u>- 9</u>	:	<u>00am</u>	= 3hrs
<u>3</u>	:	<u>00</u>	= 3hrs 00min
			+ <u>1hr 30min</u>
			<u>4hrs 30min</u>

2. A school assembly started at 8.45a.m and ended at 10.00 am. How long did it last?

=	10.00a.m	
	<u>-8.45 a.m</u>	= 9.60
	<u>1.15 a.m</u>	- <u>8 45</u>
		1.15

It lasted for 1 hr 15 min

3. Luta started walking from home to school at 7:15 a.m and reached the school at 9:15a.m. How long did it take her to reach school?

=	hrs	:	min
	9	:	15a.m
	<u>- 7</u>	:	<u>15a.m</u>
	<u>2</u>	:	<u>00</u>

She took 2 Hours.

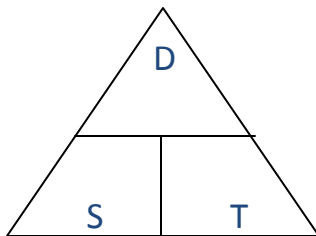
ACTIVITY

1. A mathematics lesson started at 10.50a.m and ended at 11.40a.m. How long was the lesson?

2. A church service began at 9.30am and ended at 11.00am. How long was the service?
3. We started our morning lessons at 8:05a.m and ended at 11.15a.m. How long did it take?
4. The examinatio0n started at 8:30a.m and end at 10:45a.m. How long did it take?
5. A show at the National theatre started at 8.20 lm and ended at 11.10pm.How long was the show?
6. A football match started at 8.50am and ended at 12.05p.m. How long was the match?
7. How long is it from 1:25a.m to 1:00p.m?
8. A bus travelled from Kampala at 2:00p.m and reached Kabarole at 5:20p.m. How long did it take?

FINDING DISTANCE, SPEED AND TIME

FINDING DISTANCE



To find distance, we multiply speed by time.

Distance = speed x Time

$$D = S \times T$$

Distance is commonly given in km.

1. Find the distance a driver covers in 3 hours at a speed of 90km per hours.

Distance = speed x time

$$\begin{aligned}
 \text{Distance} &= S \times T \\
 &= 90\text{km/hr} \times 3 \text{ hours} \\
 &= 90 \times 3
 \end{aligned}$$

$$\text{Distance} = 270\text{km}$$

2. A cyclist rode for 4 hours at a speed of 75km per hour. Find the distance he travelled.

$$\begin{aligned}
 \text{Distance} &= \text{speed} \times \text{time} \\
 \text{Distance} &= 75\text{km/hr} \times 4 \text{ hours} \\
 &= 300\text{km}
 \end{aligned}$$

3. A bus moved from Kampala to Mbale at a speed of 120km per hour for 5 hours.

Find the distance covered

$$\begin{aligned}
 \text{Distance} &= \text{speed} \times \text{Time} \\
 \text{Distance} &= 120\text{km} / \text{hr} \times 5 \text{ hours} \\
 &= 120 \times 5
 \end{aligned}$$

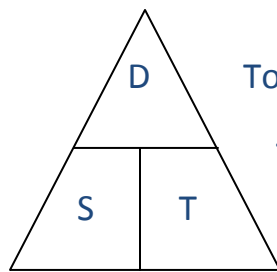
$$\text{Distance} = 600\text{km}$$

ACTIVITY

- Find the distance covered by a cyclist at 90km per hour for 3 hours.
- A school truck travelled for $2\frac{1}{2}$ hours at an average speed of 60km per hour. What distance did it cover?
- A car travelled for 6 hours at an average speed of 75km/hr. What distance did it cover?
- My father travelled in a car at a speed of 80km per hour for 7 hours.
Calculate the total distance he covered.
- A tourist moved from Kampala to Kapchorwa at an average speed of 125km.p.h. Find the distance he covered.
- Omonya moved from Masaka to Lira at a speed of 80km.p.h for $8\frac{1}{2}$ hrs. What distance did he cover?

7. A car travelled for 4 ½ hours at an average speed of 84k per hour. How long was the journey?

FINDING TIME (When distance and speed is given)



To find time (T) we divide distance by speed.

$$\text{Time} = \frac{\text{Distance}}{\text{speed}}$$

$$T = \frac{D}{S}$$

Time is always calculated in hours, minutes and seconds.

1. Calculate the time taken by a car travelling at 60k.p to cover a distance of 480km.

$$\text{Time} = \text{Distance} \div \text{speed}$$

$$\text{Time} = \frac{D}{S}$$

$$\text{Time} = \frac{480\text{km}}{60\text{ km/hr}}$$

$$\text{Time} = 8 \text{ hours}$$

2. A driver covered a journey if 240km at a speed of 80km per hour. Calculate the time he took.

$$\text{Time} = \text{Distance}$$

Speed

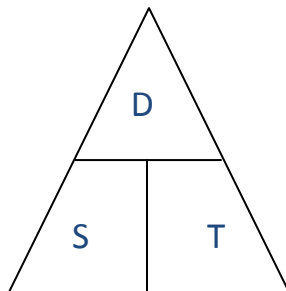
$$\begin{aligned}\text{Time} &= \frac{240\text{km}}{80\text{km/hr}} \\ &= \frac{24}{8}\end{aligned}$$

$$\text{Time} = 3 \text{ hours}$$

ACTIVITY

1. Mugoda has to cover a journey of 320km at a speed of 80km per hour. What time does he need?
2. What time is needed to cover a journey of 80km at a speed of 60km per hour?
3. Calculate the time required for a pupil to walk a distance of 6km at a speed of 2km.p.h.
4. Find the time taken for a bus to travel from Kampala to Tororo a distance of 640km at a speed of 80km.p.h.
5. Nakitto covered a journey of 128km at a speed of 64km per hour. What time did she take?
6. A bus moved from Kampala to Arua at a speed of 120km per hour to cover a distance of 600km. What time is needed to cover the distance?
7. Find the time needed to cover a distance of 960km at a speed of 20km per hour.

FINDING SPEED (When distance and time is given)



To find speed, we divide the distance by time.

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$S = \frac{D}{T}$$

Speed is calculated in km/hr, metres/min etc

1. What is the average speed of a cyclist who travelled and completed a distance of 150km in 3 hours?

$$\begin{aligned}
 \text{Speed} &= \text{Distance} \div \text{time} \\
 &= 150\text{km} \div 3\text{hrs} \\
 &= 50 \\
 &= \frac{150}{3} \\
 &= 50
 \end{aligned}$$

Speed = 50km/hr

2. A driver covered a journey of 200km in 2 hours. At what speed was he travelling?

$$\begin{aligned}
 \text{Speed} &= \text{Distance} \div \text{time} \\
 &= 200\text{km} \div 2\text{hours}
 \end{aligned}$$

$$\begin{aligned}
 &= 100 \\
 &= \frac{200}{2} \\
 &= 100
 \end{aligned}$$

Speed = 100km/hr

ACTIVITY

1. Find the speed a motorist needs to cover a distance of 360km in just 3 hours.
2. A taxi driver covered a journey of 320km in 4 hours. At what speed was he driving?
3. Find the average at which a car should driver if he is to cover 365km in 5 hours.

4. A driver covered a distance of 720km in 4 hours. Find the average speed at which he is driving.
5. Alobo travelled from Kitgum to Luwero a distance of 6/6km in 7 hours. Find the average speed at which he was driving.
6. Matsiko spent 8 hours on his way from Kampala to Masaka a distance of 560km. At what speed was he travelling?
7. A cyclist covered a distance of 60km in $2\frac{1}{2}$ hours. At what speed was he cycling?
8. An athlete covered 5km in $\frac{1}{2}$ an hour. At what speed was she running?

SIMPLE RATES AND PROPORTION

1. A book costs 500/=. What is the cost of three such books?

1 book costs shs. 500

3 books will cost 3×500

$$= 500$$

$$\times 3$$

$$1500$$

3 books will cost 1500

2. 6 pens cost shs. 900. What is the cost of 1 pen?

6pens cost 900/=

1 pen will cost 900

$$6$$

1 pen will cost 150/=

3. 5 shirts cost 5000= . Find the cost of 12 similar shirts.

5 shirts cost 5000

1 shirt will cost 5000

$$5$$

$$= 1000$$

Therefore 12 shirts will cost 12×1000

$$= 12000/=$$

4. 12 books cost 2400/=. Find the cost of 8 similar books.

12 books cost 2400/= Therefore 8 books cost 8×200

1 book costs $\frac{2400}{12}$

$$12$$

$$= 1600/=$$

$$= 200/-$$

ACTIVITY

1. The cost of 2 books is 2400/=. What is the cost of one book?
2. The cost of 1 table is 10,500/=. Find the cost of similar tables.
3. Find the cost of 15 coloured pencils if each pencil costs 250/.
4. A leap of 20 oranges cost shs.1000. Find the cost of one orange.
5. 1 kg of sugar costs 1250/= Find the cost of 5kg.
6. 3 pencils cost 450/=. What is the cost of 10 pencils?
7. 5 bags of cement weigh 250kg. What is the weigh of 4 bags of cement?
8. A dozen of dressing costs shs.28, 500, find the cost of one dress.
9. A pair of shoes costs shs.5000. How much shall 1 pay for 3 pairs.
10. 6 pens cost 1200/=. Find the cost of 15 similar pens.

SHOPPING BILLS AND BALANCES

1. A man had 500/=. He bought 2kg of sugar at 1200/= per kg.

3 bars of soap at 700/= a bar

How much balance did he have left?

Sugar	Soap	Total	Balance
-------	------	-------	---------

$1200/= \times 2$	700×3	$2400 + 2100$	$5000 - 4500$
$2400/=$	$2100/=$	4500	$500/=$

Balance left was 500=

2. A mother bought the following items from one shop.

4kg of rice at 800/= per kg.

500g ($\frac{1}{2}$ kg) of salt at 800 a kilo

1 bunch of matooke at 6000/-

If she had shs.10, 000/=-, how much change did she have left?

Rice	Salt	Matooke	Total	Balance
800×4	$800 \times \frac{1}{2}$	600×1	3200 400 6000	10,000 -9600
$3200=$	$400=$	6000	9600	400

The balance left was 400.

ACTIVITY

1. A mother had 5000= and bought the following items.

12 books at 100/= each

Geometry set at 1500/=

2 pencils at 200/= each

- Find the her total expenditure
- How much balance did she have left?

2. Amos bought the following items from the market.

5 kg of meat at shs. 2200 each

500g (1/2 kg) salt at shs.900 a kg

2 bars of soap at 1000 each

4kg of rice at 1500/= per kg

a) Calculate his total expenditure

b) Work out his balance

3. A trader bought the following items

20 pineapples at shs. 400/= each

2 tins of tomatoes at shs.6000/=

1 sack of oranges at shs. 12,000/=

a) How much money did she spend?

b) If she had 35,00/= find her balance.

5. Complete the table below.

Item	Quantity	Unit cost	Total
Chicken	2birds	Shs.3500	-
Sugar	3kg	Shs.3500	-
Rice	5kg	Shs.1000	-
bananas	2 bunches	Shs.12000	-
Total			-

6. Complete the shopping list below.

Item	Quantity	Unit cost	Total
Meat	2 ½ kg	4000/=per kg	-
Fish	6 pieces	1500/=each	-
Rice	8kg	3000/=per kg	-
Sugar	3 ¼ kg	4000=per kg	-
Omo	½ kg	Shs.3600 a kg	-
Total			-

TRANSPORT CHARGES

1. A taxi driver charges 2000/- for a trip from Kampala to Jinja per person.

- a) How much will 7 people pay for the trip?

1 person pays 2000/-

7 people will pay 7×2000

$$= 2000$$

$$\begin{array}{r} \text{ } \\ \text{ } \times 7 \\ \hline \end{array}$$

$$14,000$$

7 people will pay 14,000/=

2. A taxi charges shs.5000 for a journey from Kampala to Gayaza town.

- a) How much will 2 people pay from Kampala to Gayaza?

- b) A man boards a taxi with his wife and 3 of his grown up children to Gayaza. How much will the whole family pay?

3. It costs 5000/= for a person to travel from Kampala to Mbale. A taxi carries 14 people. How much will he collect from them?

4. It cost 2000/= to move from Jinja to Iganga.

- a) How much will 5 people pay for a journey?

- b) How much balance will 7 people receive if they had a 20,000 shilling note?

PROFIT AND LOSS

1. Sabit, bought a goat at shs.20, 000/= and later sold it at shs.25, 000. What profit did he make?

Profit = Selling price – cost price

$$= 25000/- - 20,000/-$$

$$= \text{shs.}5000$$

2. The cost of a radio is shs.100, 000. If it is sold at shs.80, 000. Find the loss made.

Loss = cost price – selling price

$$= 100,000/- - 80,000/-$$

$$= \text{shs.}20, 000/-$$

ACTIVITY

1. The cost of a pair of shoes is shs.12, 500.

A trader sold it at 15,000/=. Calculate the profit made.

2. A tailor made a dress at a cost of shs.35, 000 and sold it at shs.22, 000.

What loss did she make?

3. The cost price of an article is shs.42, 700. If I sell it at shs.50, 000. What profit do I make?

4. Batto bought a car at shs.7, 000,000 and later sold it to a friend at shs.6, 300,000. Calculate the loss made.

5. A trader buys sugar from a whole sale shop at shs.1000 per kg and sells it at shs.1200 per kg. Calculate his profit.

6. Adeke bought a suit at shs.135, 200 and sold it at shs.150, 000. How much profit did he make?

7. The cost of a radio is shs.150, 000. If it is sold at shs.127, 000. Find the loss made.

8. Musa bought a suit at shs.270, 000 it at shs.185, 000. What loss did he make?

9. Mukama bought a bag of sugar at shs.90, 000 and sold it at a reduced price of 75,000. Find the loss he made.

10. The cost of one packet of milk was 25,000. A trader sold it at shs.19, 000. Find the loss made.

TOPIC: ALGEBRA

SUBTOPICS

- Collecting like terms
- Collecting like terms and simplifying
- Forming Algebraic expressions from phrases
- Simplifying expressions by Removing brackets
- Simplifying Algebraic expressions
- Substitution (value expressions)
- Solving equations by subtracting
- Solving equations by adding
- Finding unknowns by
 - a) Squaring
 - b) Using square roots
- Finding sides of squares using square roots
- Finding the unknown sides of a figure using the perimeter
- Finding unknown sides of figure when area is given
- Finding the missing sides when volume is given

- Solving equations by dividing
- Solving fractional equations
- Forming and solving equations involving fractions
- Application of algebra

ALGEBRA

Collecting like terms

How many altogether?

1 pen + 1pen + 1pen + 1pen

1 p + 1p + 1p + 1p

4p

4pens

4b + 3b – 5b

7b – 5b

2b

5t – 9t + 7t

5t + 7t -9t

12t – 9t

3t

x+y+2x+yy

x+2x+y+4y

3x+5y

8w-3w-y

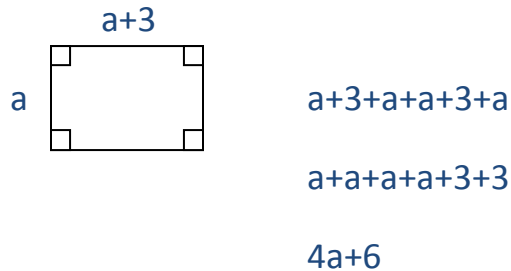
5w – y

$$9h + 3k - 4h - k$$

$$9h - 4h + 3k - k$$

$$5h + 2k$$

Find the distance around the given figures



ACTIVITY (Collecting like terms)

Work out algebraically by choosing the most suitable letter of the alphabet.

1. 2 bananas + 2 bananas
2. 4 cows + 4 cows + 10cows – 9 cows
3. 9 cats – 4 cats
4. 1 apple + 2 apples + 3 apples
5. 12 posts + 8posts – 10 posts
6. Three boys have. 5 books, 3books and 6 books respectively. How many books do they have?
7. A farmer had 13 cows. He sold off 5 cows. How many cows remained?

See: (1) Exercise 12:2 page 268 Mk Math Book 5

(2) Exercise 17.1 page 256 understanding Maths

COLLECTION *LIKE* TERMS AND SIMPLIFYING

Example

- 1) Write in short form:

$$2 \text{ balls} + 2\text{pens} + 1 \text{ ball} + 2 \text{ pens}$$

$$= (2 \text{ balls} + 1 \text{ ball}) + (2 \text{ pens} + 2 \text{ pens})$$

$$= 3 \text{ pens} + 4 \text{ pens}$$

2. Collect like terms

$$9 \text{ apples} + 4 \text{ eggs} - 5 \text{ apples}$$

$$= (9 \text{ apples} - 5 \text{ apples}) + 4 \text{ eggs}$$

$$= 4 \text{ apples} + 4 \text{ eggs.}$$

3. $a + 2b + 3a$

$$a + 2b + 3a = (a + 3a) + 2b$$

$$= 4a + 2b$$

ACTIVITY

Write in short form

1. $2 \text{ mangoes} + 3 \text{ apples} + 3 \text{ mangoes} + 1 \text{ apple}$

2. $5 \text{ cats} + 2 \text{ dogs} + 8 \text{ cats} + 6 \text{ dogs}$

3. $10 \text{ oranges} + 6 \text{ onions} - 6 \text{ oranges}$

4. $10 \text{ cows} + 8 \text{ goats} - 7 \text{ cows}$

Collect like terms and simplify where necessary:-

5. $a + 2b + 8a$

6. $4b + 6p + 5b + 2p$

7. $4b + 26b - 7b$

8. $14c - 4c - 5c$

9. $20t - 8t + 2$

10. $7y - 8m + y + 10m - 6$

FORMING ALGEBRAIC EXPRESSIONS FROM ALGEBRAIC PHRASES

Examples

<u>Phrase</u>	<u>Expression</u>
1. Add b to a	$a + b$

- | | | |
|-----|-------------------------------|-----------------------|
| 2. | Subtract b from a | $a - b$ |
| 3. | Multiply b by a | ba |
| 4. | Divide 6 by a | b/a |
| 5. | Add 5 from n | $n+5$ |
| 6. | Subtract 5 from n | $n-5$ |
| 7. | Multiply n by 5 | $n/5$ |
| 8. | Divide n by 5 | $n/5$ |
| 9. | 4 more than a | $a+4$ |
| 10. | x less than 12 | $12-x$ |
| 11. | 2 + x multiplied by 12 | $12(2+x)$ |
| 12. | Peter is 4 years older than x | $x + 4$ years |
| 13. | Double P | $2p$ |
| 14. | Tripple a | $3a$ |
| 15. | Square | x^2 |
| 16. | Half y | $y/2$ |
| 17. | $\frac{3}{4}$ of n | $\frac{3}{4}n$ |
| 18. | Average of x, b and c | $\frac{a + b + c}{3}$ |
| 19. | 2 books weigh 10 grams | $2b = 10$ |

ACTIVITY

1. A number multiplied by 3 gives 18
2. 10 less than a number is the same as 3
3. A number divided by 12 equals to 4

4. Khamis is 5 years older than Namuwenge
5. Alex is 10 years younger than Alice
6. Five boys shared shs.2500 equally
7. When P is multiplied by 2 the result is 6
8. Add 9 to a number, the result is fourteen
9. Five boys weigh 90kg
10. The sum of $2x$, x and 12 is 30
11. When a number is divided by 3 and 4 added to the result, the answer is 10.

SIMPLIFYING EXPRESSIONS BY REMOVING BRACKETS

(Understanding Maths books page 257 – 8)

1. Simplifying: $2(y+3) + 4(y+1)$

$$2(y+3)+4(y+1) \quad \longleftarrow \text{Remove brackets}$$

$$= 2xy + 2 \times 3 + 4xy + 4 \times 1$$

$$= 2y + 6 + 4y + 4$$

$$= (2y + 4y) + (6 + 4) \quad \longleftarrow \text{Collecting like terms}$$

$$= \underline{6y + 10}$$

2. Simplify: $4(t-3)+(5(2t+4)$

$$4(t-3) + 5(2t+4) = 4 \times t - 4 \times 3 + 5 \times 2t + 5 \times 4$$

$$(4t - 12) + (10t + 20) \quad \nearrow \text{(Remove brackets)}$$

$$= (4t + 10t) + (20 - 12)$$

$$= \underline{14t + 8}$$

ACTIVITY

Simplify the following expressions

1. (a) $3(m+2)+4(m-1)$ (b) $8(k-1) + 5(2k + 3)$
2. (a) $5(3n + 4) + 8(2n - 2)$ (b) $16(x+3) + 4(2x - 10)$
3. $2(8t + 5) + 3(2t - 3)$ (b) $4(5y + 6) + 3(4y - 5)$

SIMPLIFYING ALGEBRAIC EXPRESSIONS

Writing algebraic expression in their shortest forms

Example

Write in short

$$q + 7q + 4q$$

$$12q$$

$$5d + 9d - 4d$$

$$14d - 4d$$

$$10d$$

$$10x - 3x + x$$

$$7x + x$$

$$8x$$

$$6ab - 2ab$$

$$4ab$$

ACTIVITY

Write the following in short

$$p + p + p + p$$

$$4b + 3b$$

$$2m + m + 7m$$

$$5t + 7t - 9t$$

$$2b + 2b$$

$$4c + 9c - 10c$$

$$2d + 3d + 4d$$

$$a + 2a + 3a$$

$$9c - 5c - c$$

$$6x + 9x - 11x$$

SUBSTITUTION (VALUE EXPRESSIONS)

To substitute means to replace.

Example $z = 2$ and $y = 4$

a) Find the value of

i) $z + y$
 $2 + y$

$$\begin{array}{r} 2 + 4 \\ 6 \end{array}$$

(ii) $\frac{y}{z}$

$$\frac{4}{2} = 2$$

iii) $2z + 3y$
 $2xz + 3xy$
 $(2 \times 2) + (3 \times 4)$

$$\begin{array}{r} 4 + 12 \\ 16 \end{array}$$

(iv) $\frac{z}{y} + \frac{4}{z}$

v) $zy + y$
 $(zxy) + y$
 $(2 \times 4) + y$
 $8 + y = 8 + 4$
 12

$$\begin{array}{r} \frac{2}{4} + \frac{4}{2} \\ \frac{2+8}{4} \\ \frac{10}{4} \\ 2\frac{1}{2} \end{array}$$

vi) zy
 $z \times y$
 2×4
 8

(vii) $y^2 - z^2$
 $(yxy) - (zxz)$
 $(4 \times 4) - (2 \times 2)$
 $16 - 4$
 12

viii) $z^2 + y^2$
 $zxz + y \times y$
 $(2 \times 2) + (4 \times 4)$
 $4 + 16$

ACTIVITY

Given that the value of $a = 2$, $b = 3$, $C = 4$ and $v = 1\frac{1}{2}$

Find the value of the following'

1. $a + b + c$

2. $4a + 3a + ca$

3. $2a + 3c - b$

4. $4a + cv$

5. $\frac{a}{b} + b - \frac{b}{c}$

6. $b + 3ca$

7. $2a + 2b + 3c$

8. $8a - b - c$

9. $11a - b + c$

10. $16a + 2v$

11. $A \times b \times v$

12. $\frac{a}{b} + v$

13. $\frac{acb}{c}$

14. $\frac{1}{4}acb$

15. $\frac{a}{c} + \frac{b}{c}$

MORE SUBSTITUTION

If $a = 2$, $b = 4$ and $c = 6$

Evaluate the following

1. $a + b + c$

$$2 + 4 + 6$$

$$12$$

ii) abc

$$axbxc$$

$$2 \times 4 \times 6$$

$$8 \times 6$$

$$48$$

iii) $\frac{a}{b} + \frac{b}{c}$

$$\frac{2}{4} + \frac{4}{6}$$

$$6 + 8$$

$$12$$

$$\frac{14}{12}$$

$$12$$

$$1 \frac{1}{6}$$

iv) $ab + cb$

$$axb + cxb$$

$$(2 \times 4) + (6 \times 4)$$

$$32$$

v) $ac - ab$

$$axc - axb$$

$$(2 \times 6) - axb$$

$$12 - 8$$

$$4$$

vi) $\frac{ac}{b}$

$$\frac{axc}{b}$$

$$\frac{2 \times 6}{4}$$

$$\frac{12}{4}$$

$$\underline{3}$$

vii) $a + b - c$

$$2 + 4 - 6$$

$$6 - 6$$

$$0$$

viii) $2a + b + 3c$

$$2 \times a + b + 3 \times c$$

$$(2 \times 2) + 4(3 \times 6)$$

$$4 + 4 + 18$$

$$26$$

More substitution

c) If $h = 2$

Find the value of $5h$

$5h$ means $5 \times h$

$$5 \times 2$$

$$10$$

What is the value of $7d$ if $d = 6$

$7d$ means

$$7 \times d$$

$$7 \times 6$$

$$42$$

If $y = 6$

What is the value of $\frac{24}{y}$

$\frac{24}{y}$ means $24 \div y$

$$y$$

$$24 \div 6$$

$$= 4$$

If $x = 10$

What is the value of $\frac{x}{2}$?

X is the same as $x \div 2$

$$10 \div 2$$

$$= 5$$

If $y = 1$

What is the value of $8y$

$8y$ means $8 \times y$

$$8 \times 1$$

$$8$$

If $a = 12$. What is the value of:

i) $\frac{a}{3}$
 $\frac{12}{3}$
4

ii) $\frac{2a}{3}$
 $\frac{2 \times 12}{3}$
 $\frac{24}{3}$
8

iii) $\frac{1}{2}a$
 $\frac{1}{2} \times a$
 $\frac{1}{2} \times 12$
 $\frac{1 \times 12}{2} = 6$

iv) $\frac{24}{a}$
 $\frac{24}{12}$
2

ACTIVITY

If $P = 2$, $q = 4$ and $r = 8$

Evaluate the following:

1. $q + p + c =$
2. $\frac{1}{2}r + \frac{1}{4}q =$
3. $\frac{pr}{q}$
4. $\frac{pqr}{8} =$
5. $\frac{p}{q} + \frac{q}{r}$
6. $\frac{q}{r} - \frac{p}{q}$
7. $4p + q - r$
8. $\frac{1}{2}r + \frac{1}{2}q - \frac{1}{2}p =$

Solving equations by subtracting

Example

1. Solve

$$n + 5 = 11$$

$$n + 5 - 5 = 11 - 5$$

$$n = 6$$

- ii) $30 + g = 50$

$$30 - 30 + g = 50 - 30$$

$$g = 20$$

iii) Ten plus a number gives 32. What is the number?

$$10 + n = 32$$

$$10 - 10 + n = 32 - 10$$

$$n = 22$$

Therefore the number is 22

iv) What number when added to 19 gives 30 as the answer?

Let the number be y

$$19 + y = 30$$

$$19 - 19 + y = 30 - 19$$

$$y = 11$$

Therefore the number is 11

v) $n + 7 = 13$

$$n + 7 - 7 = 13 - 7$$

$$n = 6$$

vi) $a + 16 = 20$

$$a + 16 - 16 = 20 - 16$$

$$a = 4$$

vii) + 8 = 14

$$\text{input} + 8 - 8 = 14 - 8$$

$$\text{input} = 6$$

viii) $7 + \text{input} = 13$

$$7 - 7 + \text{input} = 13 - 7$$

$$\text{input} = 6$$

Solving equations by adding

Example

Solve

i) $n - 5 = 3$

$$n - 5 + 5 = 3 + 5$$

$$n = 8$$

ii) $x - 17 = 23$

$$x - 17 + 17 = 23 + 17$$

$$x = 40$$

iii) When 3 is subtracted from a number the answer is 10. What is the number?

Let the number be a

$$a - 3 = 10$$

$$a - 3 + 3 = 10 + 3$$

$$a = 13$$

Therefore the number is 13

iv) I think of a number take away 5 from it the result is 8. What is the number?

$$M - 5 = 8$$

$$M - 5 + 5 = 8 + 5$$

$$M = 13$$

v) $d - 20 = 13$

$$d - 20 + 20 = 13 + 20$$

$$d = 33$$

vi) $y - 11 = 5$

$$y - 11 + 11 = 5 + 11$$

$$y = 16$$

vii) $\square - 5 = 8$

$$\square - 5 + 5 = 8 + 5$$

$$\square = 13$$

viii) $6 - x = 14$

$$6 - 6 - x = 14 - 6$$

$$\underline{-x} = \underline{8}$$

$$\underline{-1} \quad \underline{-1}$$

$$X = -8$$

ACTIVITY

Exercise 12"9, 12:10 on page 274 and 275 MK Book 5

SOLVING MORE EQUATIONS

Example

1. Solve $2x + 5 = 17$

$$2x + 5 - 5 = 17 - 5$$

$$\frac{2x}{2} = \frac{12}{2}$$

$$x = 6$$

ii) Solve $4a + 2a + 5 = 23$

$$6a + 5 = 23$$

$$6a + 5 - 5 = 23 - 5$$

$$\frac{6a}{6} = \frac{18}{6}$$

$$a = 3$$

iii) Solve $5a - 2a - 3 - 12 = 0$

$$3a - 15 = 0$$

$$3a - 15 + 15 = 0 + 15$$

$$\frac{3a}{3} = \frac{15}{3}$$

$$a = 5$$

iv) Solve $3x - 8 = x$

$$3x - x - 8 = x - x$$

$$2x - 8 + 8 = 0 + 8$$

$$2x - 8 + 8 = 0 + 8$$

$$2x = 8$$

$$\frac{2}{2} = \frac{8}{2}$$

$$x = 4$$

FINDING THE UNKNOWN INVOLVING SQUARES

NOTE: Multiplying a number by itself gives a square.

- A number multiplied itself to get a square is the square root.
- The symbol for square root is $\sqrt{\quad}$

Example

Work out the squares of the following.

- | | | | | | | | |
|----|--------------|-----|--------------|-----|----------------|-----|--------------|
| a) | 2 | (b) | 4 | (c) | 10 | (d) | 8 |
| | 2×2 | | 4×4 | | 10×10 | | 8×8 |

Finding square roots

1. Solve $x^2 = 4$

$$\sqrt{x^2} = \sqrt{4}$$

$$\sqrt{x \times x} = \sqrt{2 \times 2}$$

$$x = 2$$

- ii) $a^2 = 144$

$$\sqrt{a^2} = \sqrt{144}$$

$$\sqrt{a \times a} = \sqrt{12 \times 12}$$

$$a = 12$$

ACTIVITY

Exercise 12:16 page 280 MK Maths books

FINDING THE SIDES OF THE SQUARE USING THE SQUARE ROOT

Example

1. The area of the square is 9cm^2

Find its sides

Side x side = Area

Let the side be y

$y \times y = \text{Area}$

$$y^2 = 9\text{cm}^2$$

$$\sqrt{y^2} = \sqrt{9\text{cm}^2}$$

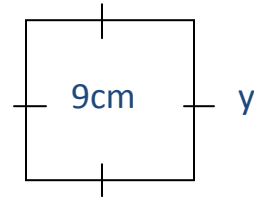
$$\sqrt{y \times y} = \sqrt{3\text{cm} \times 3\text{cm}}$$

$$y = 3\text{cm}$$

- ii) The area of the square compound is 100m^2 . Find the sides in metres.

$$\text{Side} \times \text{side} = 100\text{m}^2$$

$$52 = 100\text{m}^2$$



$$\sqrt{5 \times 5} = \sqrt{10\text{m} \times 10\text{m}}$$

$$S = 10\text{m}$$

- iii) What is the side the square whose area is 16cm^2 ?

$$S \times S = 16\text{cm}^2$$

$$52 = 16\text{cm}^2$$

$$\sqrt{5^2} = \sqrt{16\text{cm}^2}$$

$$\sqrt{S \times S} = \sqrt{4\text{cm} \times 4\text{cm}}$$

$$S = 4\text{cm}$$

FINDING THE UNKNOWN SIDE OF A FIGURE WHEN THE PERIMETER IS GIVEN

Example

1. The perimeter of a square is 36cm . Find its side in centimeters.

$$S + S + S + S = 36\text{cm}$$

$$\underline{4S} = \underline{36\text{cm}}$$

$$4 = 4$$

$$S = 9\text{cm}$$

- ii) The perimeter of a rectangle is 40cm . Its length is 15cm . Find its width.

$$L + w + L + W = \text{Perimeter}$$

$$15\text{cm} + w + 15\text{cm} + w = 40\text{cm}$$

$$15\text{cm} + 15\text{cm} + w + w = 40\text{cm}$$

$$30\text{cm} - 30\text{cm} + 2w = 40\text{cm} - 30\text{cm}$$

$$\frac{2w}{2} = \frac{10\text{cm}}{2}$$

$$w = 5\text{cm}$$

ACTIVITY

Exercise

12: 20 on page 284 – 285 MK Maths Books

Nos 1, 3, 5, 7, 9, 10 - for squares and

Nos 2, 4, 6 and 8 – for rectangle.

FINDING THE UNKNOWN SIDE WHEN THE AREA IS GIVEN

Example

The area of a rectangle is 32cm^2 . Its length is 8cm . Find its width

Length x width = Area

$$8\text{cm} \times w = 32\text{cm}^2$$

$$\frac{8\text{cm}w}{8\text{cm}} = 32\text{cm}^{2-1}$$

$$w = 4\text{cm}$$

ii) The area of a rectangular figure is 24m^2 . Its Width is 4m . Find its length.

Length x width = Area

$$L \times 4\text{m} = 24\text{m}^2$$

$4\text{m} L$

$$\frac{4\text{m}^2}{4\text{m}} = \frac{24\text{m}^{2-1}}{4\text{m}}$$

$$L = 6\text{m}$$

Therefore the length of the rectangular figure is 6metres.

ACTIVITY

Exercise 12:2 as page 286 No. 1 – 7 MK Book 5

FINDING THE MISSING SIDE WHEN THE VOLUME IS GIVEN

EXAMPLE

- i) The volume of a box is 60cm^3 . Its length is 5cm and width 4cm. Find its height.

$$L \times W \times H = \text{Volume}$$

$$5\text{cm} \times 4\text{cm} \times H = 60\text{cm}^3$$

$$\frac{20\text{cm}^2 H}{20\text{cm}^2} = \frac{60\text{cm}^{3-2}}{20\text{cm}^{2-2}}$$

$$H = 3\text{cm}$$

Therefore the height = 3cm

- ii) The volume of acuboid is 80cm^3 . Its length is 8cm and height is 2cm. What is its width?

$$L \times W \times H = \text{Volume}$$

$$8\text{cm} \times w \times 2\text{cm} = 80\text{cm}^3$$

$$16\text{cm}^2 = 80\text{cm}^{3-2}$$

$$\frac{16\text{cm}^2 W}{16\text{cm}^2} = \frac{80\text{cm}^{3-2}}{16\text{cm}^{2-2}}$$

$$\frac{16\text{cm}^2}{16\text{cm}^2} = \frac{5\text{cm}}{16\text{cm}^{2-2}}$$

Therefore the width = 5cm

ACTIVITY

Exercise 12:22 MK Mathematics Book 5 page 287

SOLVING EQUATIONS BY DIVIDING

Examples

Solve:

i) $5a = 20$

ii) $12a = 144$

Solution

i) $5a = 20$

$$\frac{5a}{5} = \frac{20}{5} \quad (\text{Divide each side by 5})$$

$$a = 4$$

ii) $12a = 144$

$$\frac{12a}{12} = \frac{144}{12} \quad (\text{Dividing each side by 12})$$

$$a = 12$$

ACTIVITY: (Exercise 12:11 Page 276 MK Maths Book 5)

Solve for the unknown:

1. $4x = 16$

2. $3x = 24$

3. $11x = 165$

4. $15n = 225$

5. $37 = y = 108$

WORD PROBLEMS INVOLVING SOLVING EQUATIONS BY DIVIDING

Examples

1. The length of a rectangle is 9cm and the width is w cm. If its area is 72cm^2

$$L \times w = \text{Area}$$

$$9 \times w = 72\text{cm}^2$$

$$9w = 72\text{cm}^2$$

$$\frac{1}{9} = \frac{8}{9}$$

$$\frac{\cancel{9}w}{\cancel{9}1} = \frac{\cancel{72}\text{cm} \times \text{cm}}{\cancel{9}\text{cm} \ 1}$$

$$\cancel{9}1 \quad \cancel{9}\text{cm} \ 1$$

$$w = 8\text{cm}$$

Therefore the width of the rectangle is 8cm.

ACTIVITY (Exercise 12:12 page 276 MK Maths Bk 5)

1. Cherimo collected 40 mangoes in 5 baskets. How many mangoes were in each basket?
2. What number when multiplied by 2 gives the answer as 30?
3. A pen costs 2p/= and a book costs shs.p. If the total cost of the books and pens is shs.300, find the cost of the book.

ACTIVITY 2: (Exercise 12:13 on page 277 MK Book 5)

Nos 1 – 25

SOLVING EQUATIONS INVOLVING FRACTIONS

Examples

1. Solve: $\frac{x}{3} = 4$

$$\frac{x}{3} = 4$$

(Multiply each side by 3)

$$3 \times \frac{x}{3} = 4 \times 3$$

$$\frac{3}{x} = 12$$

ACTIVITY

a) Find the value of the letters:

i) $\frac{a}{4} = 9$

ii) $\frac{b}{5} = 4$

iii) $\frac{m}{7} = 8$

b)

i) $\frac{2p}{3} = 4$

ii) $\frac{4b}{5} = 4$

iii) $\frac{3m}{4} = 12$
3

iv) $6 = \frac{e}{7}$

v) $\frac{p}{10} = 12$

vi) $20 = \frac{n}{10}$

FORMING AND SOLVING MORE EQUATIONS INVOLVING FRACTIONS

Examples

1. What number when divided by 4 gives 3?

Let n be the number.

$$n = 3$$

$$\frac{n}{4} = \frac{3}{1} \quad (\text{The LCD of 4 and 1} = 4)$$

$$\frac{4 \times n}{4} = \frac{3 \times 4}{1} \quad (\text{Multiply each side by the LCD}=4)$$

$$n = 12$$

Therefore the number is 12

2. A man divided his money among his 3 children and each got shs.450.

How much money did he give out?

Let the amount of money he had be m.

Each child got ----- ($\frac{m}{3}$)

$$\frac{m}{3} = 450$$

$$\frac{m}{3} = \frac{450}{1} \quad (\text{LCD of 3 and 1} = 3)$$

$$3 \times \frac{m}{3} = \frac{450}{1} \times 3$$

$$m = 1350$$

Therefore the man gave out 1,350/-

ACTIVITY

Express each of the below statements as equations and solve:

1. A father divided some money between two children and each one got shs.150. How much did the father give out?
2. What number when divide by 7 gives 6?
3. When shs.x is shared by 6 boys each gets shs.150. Find the value of x.

4. The teacher gave exercise books to 25 pupils and each one got 10 books.

How many books did the teacher have?

Ref: Exercise 12:19 on page 283 MK Maths Book 5

APPLICATION OF ALGEBRA

Examples

1. Wamala had some books. He got 3 books. Altogether he had 7 books.
How many books did he have before?

Let the books be m

He got 3 more books

He had $m + 3$ books

Altogether $m + 3 = 7$

$$m + 3 - 3 = 7 - 3$$

$$m = 4$$

He had 4 books

- ii) Alex had some goats. When he sold 5 of them he remained with 9 goats.
How many goats had he before?

Let the number of goats he had be g .

He sold 5 goats

Goats left are $g - 5$

Therefore $g - 5 = 9$

$$g - 5 + 5 = 9 + 5$$

$$g = 14$$

Alex had 14 goats

- iii) There are 4 groups in a class. Each group has the same number of pupils. Altogether there are 40 pupils. How many pupils are in each group?

Let the pupils in each group be c

There are 4 groups

Altogether = $4xc$ pupils

$$4c = 40$$

$$4 = 4$$

$$c = 10$$

Each group has 10 pupils

- iv) Alice had some balls; she divided them into 4 groups. If there were 12 balls in each group, how many balls did she have altogether?

Let the balls be b

She made 4 groups

This is $b \div 4$ groups

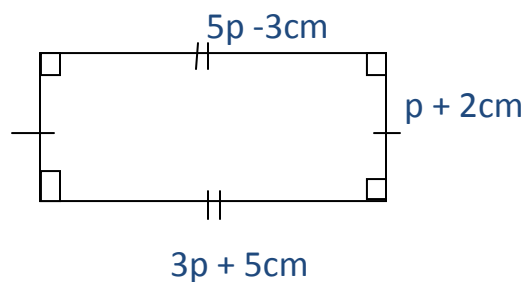
Therefore $b \div 4 = 12$

$$b \div 4 \times 4 = 12 \times 4$$

$$b = 48$$

Therefore the balls were 48

- v) Find the value of P in centimeters of the figure below.



Opposite sides of a rectangle are equal

$$5p - 3\text{cm} = 3p + 5\text{cm}$$

$$5p - 3p - 3\text{cm} = 3p - 3p + 5\text{cm}$$

$$2p - 3\text{cm} + 3\text{cm} = 5\text{cm} + 3\text{cm}$$

$$2p - 3\text{c} + 3\text{cm} = 5\text{cm} + 3\text{cm}$$

$$\frac{2p}{2} = \frac{8\text{cm}}{2}$$

$$p = 4\text{cm}$$

a) Find the actual length of the figure.

$$5p - 3\text{cm}$$

$$5 \times p - 3\text{cm}$$

$$5 \times 4\text{cm} - 3\text{cm}$$

$$20\text{cm} - 3\text{cm}$$

$$17\text{cm}$$

b) Work out the actual width of the figure

$$p + 2\text{cm}$$

$$4\text{cm} + 2\text{cm}$$

$$= 6\text{cm}$$

c) Calculate the perimeter of the figure

$$p = 2(L + W)$$

$$= 2L + 2W$$

$$2 \times L + 2 \times w$$

$$(2 \times 17\text{cm}) + (2 \times 6\text{cm})$$

$$34\text{cm} + 12\text{cm}$$

$$= 46\text{cm}$$

d) Find the area of the figure

$$A = L \times W$$

$$= 17\text{cm} \times 6\text{cm}$$

$$= 17 \times 6 \text{ cm} \times \text{cm}$$

$$= 102\text{cm}^2$$

Ref: Understanding Mathematics Book 5 Page 263 to 264)

ACTIVITY (Understanding Mathematics pupils book 5 pages 263 – 4)

1. The length of a rectangle is $2x + 6$ and the breadth is 9cm.

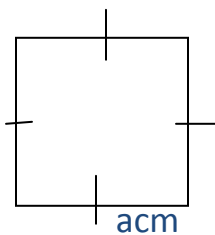
a) Write an expression for the area

b) Write the expression for perimeter

c) Simplify the expression in (a) above.

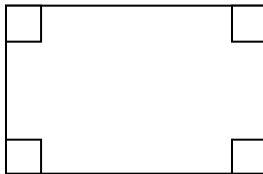
d) If the area of the rectangle is 108cm^2 . Find the value of x .

2.



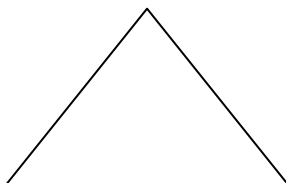
Find the perimeter

3.



The length of rectangle ABCD is 8cm and width is 5cm

Find the area



4.

— —

|

a) Find the perimeter of the given triangle

b) If the perimeter is 21cm, find the value of x .

5. If the perimeter of a square is given by the formula $4L = 32$, find the value of L .

6. The length of a rectangle is $(5x - 3)$ cm and the width is $(3x + 7)$ cm.

a) Write the expression for the perimeter of the rectangle.

b) Simplify the expression in (a) above

c) If the perimeter is 40cm, find the value of x